

Architecture Determines Observability of Transformers

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ABSTRACT

Autoregressive transformers make confident errors that output-confidence monitoring cannot catch. Activation monitors catch them only when training leaves a decision-quality signal beyond what the output already exposes. This signal is an architectural property of the trained model, fixed upstream of any monitor. Controlling for output confidence removes 60.3% of the raw activation-probe signal on average across 14 models. Raw probe signal is mostly output confidence, and output-side readouts cannot recover the residual. What remains depends on architecture and training. In Pythia’s controlled training, both matched-width configurations form the signal early. One preserves it through convergence while another erases it as perplexity continues to improve. Capability and observability are not inherently in tension. Across independently trained families this pattern persists, even as the collapse point shifts. Where the signal survives, monitoring catches what confidence cannot. On downstream QA, a WikiText-trained probe with no task-specific tuning catches about one in eight confident errors that output-confidence monitoring misses, at a 20% flag rate. These results establish signal engineering as a training-time design axis alongside loss and capability. Architecture sets the conditions for observability, and training determines what remains readable.

1 INTRODUCTION

Activation monitors for trained models operate on the premise that the target signal survives training (Kramár et al., 2026; Zou et al., 2023). Hidden states encode signals not directly exposed by outputs, including latent knowledge, geometric structure of truthful representations, and verification of generated reasoning (Burns et al., 2023; Marks and Tegmark, 2024; Zhang et al., 2025). This paper tests two output-side signals: max-softmax probability and trained final-layer predictors. A linear probe on frozen mid-layer activations (Alain and Bengio, 2017; Hewitt and Liang, 2019; Belinkov, 2022) can read a different signal: per-token decision quality beyond what output confidence explains. On some architectures, this signal is strong and stable across seeds. On others, no measured layer preserves a healthy-range, linearly readable signal.

Monitorability is therefore an architectural property before it is a monitor-design problem. We make this property measurable by defining observability: the confidence-controlled linear readability of per-token decision quality from frozen mid-layer activations (Section 2). For this decision-quality target, much of what raw probes find in hidden states is output confidence in disguise. On GPT-2 124M, raw Spearman correlation between a linear probe and per-token loss is 0.514, but only 0.289 survives after controlling for max-softmax and activation norm. Across 14 models in 6 families, controlling for confidence removes on average 60.3% of the raw signal. Raw probe scores therefore substantially overstate what activations reveal beyond output confidence.

Figure 1 previews the resulting within-family cliffs under this confidence-controlled metric.

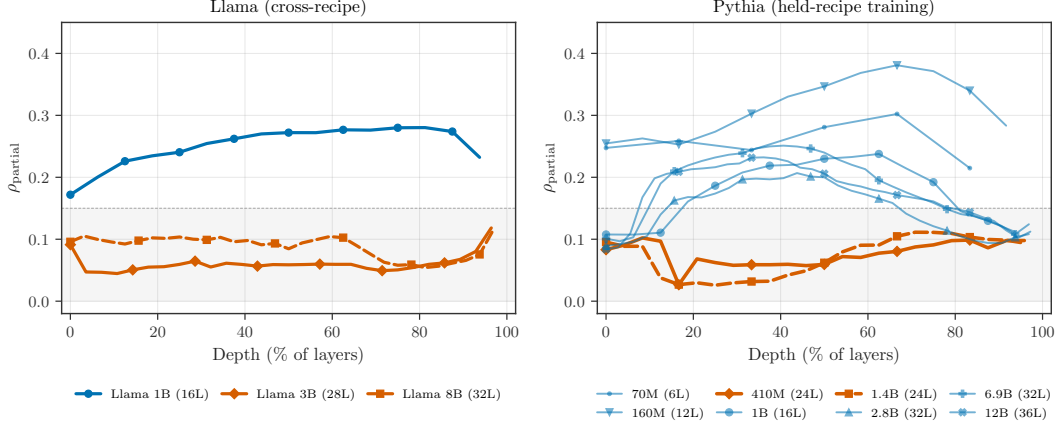


Figure 1: Within Pythia’s controlled suite (right), configuration determines observability; the same within-family discontinuity appears observationally in Llama (left, cross-recipe). The y-axis is ρ_{partial} , the partial Spearman correlation between a linear probe and per-token loss after controlling for max-softmax probability and activation norm, as defined in Section 2. Left, Llama: 1B preserves the signal, while 3B and 8B do not. Right, Pythia controlled suite: configurations outside the 24-layer, 16-head class preserve the signal, while 410M and 1.4B at this configuration collapse across a $3.5\times$ parameter gap with hidden dimensions 1024 and 2048. The 1.4B-deduped replication appears in Table 4. Same protocol, same token budget per hidden dimension. Collapse is discrete, not gradual.

The deeper question is whether the surviving signal is output-independent. We therefore introduce an output-controlled residual, r_{OC} , which additionally controls for what a trained predictor on the final-layer representation can recover. Positive r_{OC} means the mid-layer observer reads decision-quality information not recovered by the tested output-side readout. Across 26 tested models, r_{OC} tracks the confidence-controlled probe: where observability collapses, the output-controlled residual collapses with it.

Pythia supplies the controlled test for observability collapse. Three runs sharing the (24-layer, 16-head) configuration class collapse to $\rho_{\text{partial}} \approx 0.10$ despite $3.5\times$ variation in parameters and two Pile variants, while six other configurations produce ρ_{partial} between 0.21 and 0.38. Because the output-controlled residual collapses with observability, what is lost is the output-independent component isolated by the protocol. Checkpoint dynamics show a loss-monitorability dissociation. Both matched-width configurations form the signal at the earliest measured checkpoint and improve in perplexity. Training erases observability in the (24-layer, 16-head) class while the healthy configuration recovers. Within this held recipe, architecture does not prevent the signal from appearing. It determines whether training preserves or erases it. Signal engineering (Definition 3) makes signal preservation or amplification a training objective alongside loss.

Cross-family evidence extends the pattern observationally. At matched 3B scale, Qwen 2.5 and Llama differ by $2.9\times$ with non-overlapping probe-seed distributions, while Mistral 7B v0.3 and Llama 3.1 8B share the same high-level 32-layer, 32-head, 4096-hidden shape yet diverge. These comparisons are observational: within a held recipe, configuration determines preservation; across recipes, recipe choice shifts where collapse occurs. A WikiText-trained observer transfers to downstream QA without task-specific training, catching about one in eight errors missed by output-confidence monitoring at a 20% flag rate.

Table 1 summarizes the evidence hierarchy: measurement hardening, controlled within-recipe evidence, observational cross-recipe extension, and scoped downstream utility. Each tier points to the same upstream property. Architecture selection is a monitoring decision.

Table 1: Evidence hierarchy. The first three rows establish the measurement as strong and hardened. The remaining rows separate controlled within-recipe evidence from observational cross-recipe extensions and scoped downstream utility.

Claim	Evidence	Evidence type
Raw probes are confidence-contaminated	14-model confidence-control analysis	Cross-model
Confidence-independent signal exists	20-seed hardening, shuffle/random tests	Cross-seed + null
Signal is output-independent	r_{OC} , output-side MLP width sweep	Multi-method
Configuration determines observability	Pythia controlled suite (3 replications)	Controlled within recipe
Collapse is training-emergent	Pythia 1B vs. 1.4B checkpoint trajectories	Controlled within recipe
Phenomenon recurs across recipes	Llama/Qwen/Mistral/Phi/Gemma	Observational
Recipes shift which configurations collapse	Mistral 7B vs. Llama 3.1 8B at matched 32L/32H/4096d shape	Observational
Operational catch on downstream tasks	LM/SQuAD/MedQA/TruthfulQA	Scoped utility

2 METHOD

Standard probe evaluations mix decision quality with output confidence. Our method isolates the confidence residual: a linear probe on frozen transformer activations predicts a confidence-residualized binary target. All transformer weights are frozen throughout. Only the linear observer and the output-side MLP control are trained.

Definition 1 (Observability). *The observability of a transformer at layer l , equivalently its linear readability, is the degree to which a linear probe on $\mathbf{h}^{(l)}$ predicts per-token loss beyond what max-softmax probability and activation norm explain. The scalar estimator is ρ_{partial} (partial Spearman correlation controlling for these two covariates); the output-controlled residual statistic, r_{OC} , is the partial Spearman correlation obtained after additionally controlling for a trained predictor on final-layer activations. “Decision quality” refers to this confidence-residual loss signal, not to a broader notion of reasoning correctness.*

Definition 2 (Observability collapse). *An architectural configuration undergoes observability collapse under a training recipe when ρ_{partial} falls to the detection floor (Section 3) and the output-controlled residual r_{OC} approaches zero, so that linearly readable, output-independent decision quality is not preserved at any measured layer. This is a collapse of monitorable signal, not of predictive capability. Checkpoint trajectories distinguish training-time erasure from static non-formation (Section 4).*

Definition 3 (Signal engineering). *Signal engineering is the design of training procedures that preserve or amplify specified target signals as readable from activations of the frozen trained model. Whereas activation monitoring takes a frozen model and asks what can be read, signal engineering takes a target signal and asks what training preserves or amplifies it.*

Observability measures what a linear readout can extract, not what the model’s own computation uses. We use “monitorability” as the broader operational property monitors depend on; “observability” is the linearly-readable instance this paper measures. The confidence covariates test whether the mid-layer signal survives output confidence at generation time; the output-side readout tests whether it survives what a trained final-layer predictor can recover.

Scope and assumptions. We study frozen autoregressive transformers, per-token cross-entropy loss, max-softmax and activation-norm covariates, linear mid-layer observers, and output-side MLP readouts. This is a deliberately narrow definition of decision-quality observability, not a claim about reasoning correctness or semantic truth. The restriction is the point. If a weak, single-pass linear observer reads a signal that output-side predictors miss, the model preserved monitorable information.

If even nonlinear probes and full layer sweeps do not recover it, the architecture-recipe did not preserve that signal under this protocol.

Target construction. For each non-padding token position i in the evaluation corpus, we compute per-token cross-entropy loss ℓ_i , max-softmax probability $\hat{p}_{\max,i}$, and $\|\mathbf{h}_i\|$, the L2 norm of the residual stream at layer l (block output; equivalently the input to layer $l + 1$). An ordinary least squares (OLS) regression $\ell_i \sim \hat{p}_{\max,i} + \|\mathbf{h}_i\|$ fitted on the training split produces residuals $\epsilon_i = \ell_i - \hat{\ell}_i$. The same fitted coefficients are applied without refitting to validation and test tokens, so the residual target on held-out data uses train-fit residualizer coefficients. The binary target is:

$$y_{\text{resid},i} = \mathbf{1}[\epsilon_i > 0]. \quad (1)$$

The target is approximately balanced: OLS residuals are mean-zero by construction and empirically near-symmetric across WikiText-103 tokens. Binary cross-entropy supervision rather than regression on the residual magnitude is a design choice analyzed in [Section 3](#). Because $\|\mathbf{h}_i\|$ is layer-specific, the binary target is constructed separately for each model and layer. Layer profiles therefore measure the readable confidence residual signal at each depth under that layer’s own norm control.

Observer probe. The observer is a linear head on frozen residual stream activations at layer l :

$$o(\mathbf{h}^{(l)}) = \mathbf{w}^\top \mathbf{h}^{(l)} + b. \quad (2)$$

Training uses binary cross-entropy on y_{resid} with Adam (lr = 10^{-3} , batch size 4096, weight decay 10^{-4}) for 20 epochs. The probe has $d + 1$ parameters. No activation function, no hidden layer.

Evaluation: partial correlation. All reported correlations are Spearman rank partial correlations controlling for confounds $\mathbf{c} = [\hat{p}_{\max}, \|\mathbf{h}\|]$. After rank-transforming all variables, we project out the covariates:

$$\begin{aligned} \rho_{\text{partial}}(s_\theta, \ell \mid \mathbf{c}) &= \text{corr}(r_{s_\theta|\mathbf{c}}, r_{\ell|\mathbf{c}}) \\ r_{x|\mathbf{c}} &= x - \mathbf{C}(\mathbf{C}^\top \mathbf{C})^{-1} \mathbf{C}^\top x. \end{aligned} \quad (3)$$

Here $s_\theta = o(\mathbf{h}^{(l)})$ is the probe score and $\mathbf{C}$ is the matrix of ranked covariates with an intercept column. Spearman correlation is invariant to monotone transformations of a scalar score. Logit-level temperature scaling can reorder max-softmax values across examples in multiclass settings, so the nonlinear MLP and logit-entropy checks in [Section 3](#) serve as the empirical robustness check.

Output-controlled residual. An MLP ($d \rightarrow 64 \rightarrow 1$, ReLU) trained on last-layer activations to predict loss $\hat{\ell}_\psi^{(L)}$ is the output-side control. The MLP is trained with Adam (lr = 10^{-3} , batch size 1024, weight decay 10^{-4}) for 20 epochs on the probe’s training split; its predictions on the held-out evaluation tokens form $\hat{\ell}_\psi^{(L)}$:

$$r_{\text{OC}} = \rho_{\text{partial}}\left(s_\theta^{(l)}, \ell \mid \mathbf{c}, \hat{\ell}_\psi^{(L)}\right). \quad (4)$$

Positive r_{OC} means the mid-layer observer reads information not recovered by the trained final-layer predictor. A width sweep (64–512 units) tests whether this result depends on predictor capacity ([Section 3](#)). [Table 2](#) maps each monitoring signal to the questions it can answer.

Each level of this hierarchy tests against a different alternative explanation. Confidence control tests whether the probe merely recapitulates output confidence. Output control tests whether the mid-layer signal is merely re-reading information recoverable from the final-layer representation by the tested output-side predictors. The observer is complementary to those predictors when r_{OC} is positive; when r_{OC} collapses together with ρ_{partial} , this complementary, output-independent component is no longer preserved in linearly readable form under this protocol. Under a held recipe, this is what it means for architecture to determine observability: it determines whether the output-independent component that activation monitors need survives. We use $\rho_{\text{partial}} \leq 0.15$ as the empirical detection floor because random, shuffled, and underpowered probes cluster near or below this range ([Section 3](#)). Within the Pythia controlled suite, healthy configurations begin at $\rho_{\text{partial}} \geq 0.21$ (exact minimum 0.208, Pythia 2.8B), with the gap between these thresholds treated as an indeterminate margin within that suite.

Table 2: What each monitoring signal establishes. The output-controlled residual r_{OC} is the only single-signal test of output-independence.

Signal / protocol	Predicts loss	Removes confidence	Tests output independence	Detects collapse
Output confidence	✓			
Raw activation probe	✓			Weakly
ρ_{partial} (conf.-controlled)	✓	✓		Partial
r_{OC} (output-controlled)	✓	✓	✓	✓
Nonlinear / layer sweeps	✓	✓	Partly	Artifact check
Pythia controlled suite	✓	✓	✓	Controlled (within recipe)

Seed agreement. Seed agreement τ_{seed} is the mean pairwise Spearman correlation of observer scores across k independently initialized probes on the same frozen activations:

$$\tau_{\text{seed}} = \frac{2}{k(k-1)} \sum_{i < j} \rho_s(o_i, o_j). \quad (5)$$

Observed τ_{seed} on frozen activations typically exceeds 0.9, consistent with convergence to a single direction across initializations. Full seed-count analysis in [Section 3](#).

Target validity. We construct the target by residualizing against confidence, then evaluate against the same covariates. Six empirical tests address this circularity, organized as one construction test, three selectivity tests, and two null-distribution tests (full results in [Section 3](#)). The construction test confirms that residualizer fitting does not leak: a separate residualizer-fit split preserves the regime on four representative models. Selectivity tests confirm that the target is not trivially recoverable: hand-designed activation statistics fail under the same target, the probe fails on Llama 3B under the identical target, and training the probe on C4 web text under the same target-construction protocol fails. Null-distribution tests confirm that the signal is not a probing artifact: random untrained probes and a shuffle test on randomized labels both produce near-zero ρ_{partial} .

Evaluation data. Core observability probes train and evaluate on WikiText-103 ([Merity et al., 2017](#)) with standard splits. Probes train on the train split. Layer selection is two-stage: a seed-42 sweep produces a candidate set (the top four layers by seed-42 ρ_{partial} , depth-filtered to $\leq 80\%$ of total depth, plus the seed-42 argmax), and the reported layer is the argmax of the 7-seed mean (seeds 43–49) over candidates. Reported ρ_{partial} is the 7-seed distribution at that layer; the top-three layer flatness in [Appendix B](#) bounds selection-induced optimism. The standard token budget is 350 examples per hidden dimension (ex/dim), where an example is one non-padding token position. [Section 3](#) reports 20-seed hardening on GPT-2 124M, [Section 5](#) reports frozen-probe transfer to downstream tasks, [Appendix G](#) reports cross-domain transfer to C4 and other corpora, and [Appendix B](#) reports test-split confirmation and layer-selection sensitivity. Because the observer is a linear direction in a d -dimensional residual stream, cross-scale comparisons require token budgets scaled by hidden dimension. On Qwen 2.5 0.5B ($d = 896$), a seven-point sweep from 150 to 1,000 ex/dim reveals a budget threshold between 450 and 600 ex/dim: below 450, ρ_{partial} is precise but near the detection floor; above 600, the signal rises to the reported healthy value ([Appendix F](#)). Low-budget estimates have low seed variance, so small confidence intervals do not guarantee adequate power. All reported cross-scale comparisons use budgets above each model’s empirically adequate threshold. Collapsed models (Llama 3B, 8B; Pythia 410M, 1.4B) remain collapsed at absolute token budgets well above the detection regime, so underpowering does not explain collapse.

Statistical model. Cross-family comparisons use a mixed-effects model with per-seed observations, random intercepts per model (absorbing within-model correlation from shared data and layer selection), and fixed effects for $\log_{10}(\text{params})$ and family:

$$\rho_{\text{partial } ij} \sim \log_{10}(N_i) + \text{family}_i + (1 \mid \text{model}_i). \quad (6)$$

Full specification and variance decomposition in [Section 5](#). Family-level significance is assessed via a Monte Carlo permutation test (50,000 samples) on model-mean ρ_{partial} , with $\log_{10}(\text{params})$ residualized before permutation ([Section 5](#)). The protocol isolates per-token decision quality from output confidence: the residual signal it measures is what activations expose beyond the tested output-side readouts.

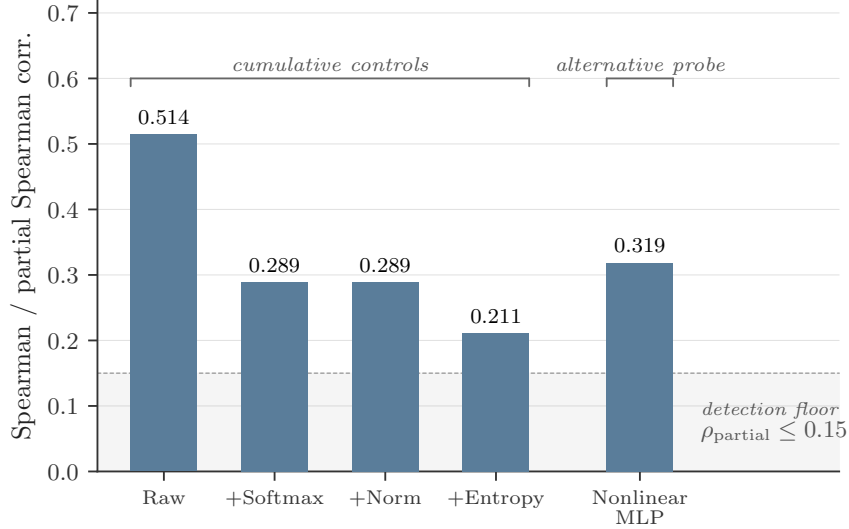


Figure 2: Control sensitivity on GPT-2 124M. The first four bars are cumulative: raw Spearman (0.514), then adding max-softmax control (0.289), then norm (0.289), then logit entropy (0.211). The fifth bar is an independent nonlinear MLP control (0.319), confirming the residual is not a nonlinear function of max-softmax confidence and activation norm.

3 ISOLATING A MONITORABLE DECISION-QUALITY SIGNAL

Before observability can serve as an architecture-and-training design axis, the signal it measures must be separated from output confidence, target-construction artifacts, and information already recoverable by tested output-side readouts. This section isolates that signal on WikiText-103, using GPT-2 (Radford et al., 2019) as the hardening case and cross-model tests where needed. The goal is not measurement for its own sake: it is to identify a monitorable decision-quality component whose preservation can then be compared across architectures and training recipes.

Raw probes are confidence-contaminated. On GPT-2 124M, raw Spearman correlation between a linear probe and per-token loss is 0.514, but only 0.289 survives after controlling for max-softmax and activation norm. Across 14 models in 6 families, controlling for confidence removes $60.3\% \pm 7.0\%$ of the raw signal, ranging 43.8%–75.8% across the cohort (per-model breakdown in Appendix C). On this decision-quality target, half to three-quarters of what probes find in hidden states is output confidence in disguise.

Two additional robustness checks do not eliminate the residual signal (Figure 2). A nonlinear MLP trained on $[\hat{p}_{\text{max}}, \|\mathbf{h}\|]$ to predict loss does not reduce the residual signal: $\rho_{\text{partial}} = 0.319$, compared with 0.289 under the linear standard control. The nonlinear MLP can represent Platt scaling and other scalar recalibrations of max-softmax confidence as special cases, so the signal is not explained by miscalibrated confidence. Logit entropy as a third covariate absorbs a further 27% ($\rho_{\text{partial}} = 0.211$). The observer partially reads the shape of the output distribution.

Raw correlations are easy to produce. Survival after controlling for confidence is the bar.

A learned linear observer recovers a stable residual signal. Every hand-designed activation statistic loses signal after controlling for confidence. On GPT-2 124M, Forward-Forward goodness, active ratio, activation entropy, and activation norm all produce near-zero or negative ρ_{partial} after controlling for confidence and activation norm (Table 3).

A linear probe trained on the residualized binary target (Equation (1)) recovers $\rho_{\text{partial}} = 0.282 \pm 0.001$ on frozen GPT-2 124M (20 seeds at layer 11, the hardening reference layer; Equations (2) and (3)) with seed agreement 0.993 (Equation (5)). The standard 7-seed protocol at the peak layer 8 yields 0.288. Random untrained probes produce near-zero ρ_{partial} on tested architectures (Table 6). Binary supervision produces higher seed agreement than regression because the heads converge on a shared decision boundary (Appendix C).

Table 3: Hand-designed activation statistics on GPT-2 124M (layer 11). All collapse to near-zero partial correlation under confidence controls.

Observer	Partial corr
FF goodness ($\sum h_i^2$)	-0.010
Active ratio	-0.057
Activation entropy	-0.110
Activation norm	-0.002
Learned linear binary	+0.282

Validity checks address target and probe artifacts. The target construction residualizes against confidence and the evaluation applies the same covariates (Section 2). Six tests in three categories address the resulting circularity: a construction test, three selectivity tests, and two null-distribution tests.

The construction test confirms that residualizer fitting does not leak. Fitting the OLS residualizer on a disjoint document pool and applying those coefficients without refit to the probe-training and evaluation pools preserves the healthy/collapsed regimes on 4 representative models, with split-fit ρ_{partial} differing from the same-pool baseline by at most 0.017 (Appendix B.2). Selectivity tests confirm that the target is not trivially recoverable: hand-designed activation statistics fail (above), the observer fails on Llama 3.2 3B at 0.091 under the identical target, and training the observer on C4 web text under the same target-construction protocol fails on Qwen 2.5 7B at $\rho_{\text{partial}} = -0.029$ (Appendix G). Null-distribution tests confirm that the signal is not a probing artifact: random untrained probes produce near-zero ρ_{partial} across tested architectures, and a probe trained on shuffled labels yields $\rho_{\text{partial}} = 0.014 \pm 0.019$ across 10 permutations on GPT-2 124M, against a real probe score of 0.292 that exceeds every shuffled permutation.

The residual is partly output-independent. Observability exists at every GPT-2 124M layer, peaking at layer 8 of 12 ($\rho_{\text{partial}} = 0.288$). The output-side control absorbs about half of the layer 8 signal, but $r_{\text{OC}} = 0.140 \pm 0.008$ survives (3 seeds; Equation (4)). Under this output-side control, the observer is not merely re-reading final-layer information from an earlier layer. It reads something the tested output-side predictors do not recover.

A width sweep (64–512 units) on the output-side MLP applied to Qwen 2.5 7B leaves the output-controlled residual stable at every width: capacity is not the bottleneck. A 30-resample document-level bootstrap on Qwen 2.5 7B yields $\rho_{\text{partial}} = 0.238$ (95% CI [0.215, 0.270]), stable under data resampling. A matched two-layer MLP probe is statistically equivalent to the linear probe across 10 models (TOST at margin ± 0.03 ; Appendix A.4). In Llama 3.2 3B, a swept-HP nonlinear probe with held-out hyperparameter selection scores 0.102, well below the healthy floor: a weak linear readout does not explain the later collapse results. The observer direction lies in a low-variance subspace that variance-dominant and reconstruction-trained decompositions partially obscure (Appendix D).

Section 4 identifies under controlled training the architectures that preserve this output-independent component and those that erase it.

4 PYTHIA: A CONTROLLED WITHIN-RECIPE TEST

Pythia supplies the controlled architecture test. Holding corpus, tokenizer, optimizer family, and schedule template fixed, the eight-size suite varies architecture across scale. Under this held recipe, every tested run sharing the 24-layer, 16-head configuration class collapses to $\rho_{\text{partial}} \approx 0.10$, while six other configurations occupy a separated healthy band (0.21 to 0.38). We refer to this discrete loss of readable decision-quality signal as *observability collapse*. This section establishes the controlled separation, rules out layer-selection, readout-capacity, and measurement-artifact alternatives, and shows that collapse is training-emergent rather than a static representational incapacity.

The Pythia suite (Biderman et al., 2023) holds corpus, tokenizer, optimizer family, and schedule template constant across eight sizes from 70M to 12B. Learning rate and batch size scale with model size while architecture varies. The three 24-layer, 16-head collapse points are Pythia 410M (0.105), Pythia 1.4B (0.106), and Pythia 1.4B-deduped (0.100, same architecture as 1.4B but trained on the

deduplicated Pile (Gao et al., 2021) variant), each measured across 7 probe seeds. The three fall within 0.006 of each other despite $3.5\times$ variation in parameters, $2\times$ in hidden dimension, $2\times$ in head dimension, and two Pile variants. At 410M the seed variance is ± 0.0009 , the tightest measurement in the paper.

Six other Pythia configurations, spanning depths 6–36 and head counts 8–40, produce ρ_{partial} between 0.21 and 0.38. Pythia 1B (16-layer, 8-head, 0.246) sits between the two collapsed models in parameter count but produces healthy observability, ruling out the 400M–1.5B size band as the collapse trigger. The collapse is tied to the (24-layer, 16-head) configuration class, not to scale.

Table 4: Pythia suite under held-recipe training (The Pile, same tokenizer, same optimizer family). Three runs sharing the (24 layers, 16 heads) configuration class collapse (bold), replicated across a $3.5\times$ parameter gap (410M vs. 1.4B), $2\times$ hidden dimension and $2\times$ head dimension, and two Pile variants (regular vs. deduplicated at 1.4B). Six other configurations produce ρ_{partial} between +0.21 and +0.38. The output-controlled residual r_{OC} collapses at the same three runs (near zero or negative), while healthy configurations range +0.09 to +0.17. All values are 7-seed means on held-out seeds 43–49 (layer selected via seed 42). Standard deviations are across seeds.

Model	Layers	Heads	Hidden	Head_dim	ρ_{partial}	Seed std	r_{OC}
Pythia-70M	6	8	512	64	0.301	0.001	0.147
Pythia-160M	12	12	768	64	0.382	0.004	0.169
Pythia-410M	24	16	1024	64	0.105	0.001	0.043
Pythia-1B	16	8	2048	256	0.246	0.012	0.120
Pythia-1.4B	24	16	2048	128	0.106	0.006	0.003
Pythia-1.4B-deduped	24	16	2048	128	0.100	0.007	−0.012
Pythia-2.8B	32	32	2560	80	0.208	0.007	0.088
Pythia-6.9B	32	32	4096	128	0.240	0.012	0.110
Pythia-12B	36	40	5120	128	0.238	0.005	0.097

The right panel of Figure 1 shows layer-wise profiles for all eight Pythia sizes: six configurations peak at mid-to-late depth between 0.21 and 0.38, while the two 24-layer, 16-head configurations (410M and 1.4B) stay flat near 0.10 across every depth. The deduplicated 1.4B replication confirms the same pattern (Table 4). At the final checkpoint, the collapsed configurations show no mid-layer peak. The measured residual-stream states show no layer at which healthy-range observability remains. The left panel shows Llama under a different training recipe, reproducing the same within-family discontinuity at a different configuration. Llama 3.2 1B is healthy at (16-layer, 32-head), while 3.2 3B and 3.1 8B collapse at (28-layer, 24-head) and (32-layer, 32-head).

The three replications form three controlled comparisons, two holding the (24-layer, 16-head) class constant while varying width or corpus, and one holding the Pythia recipe constant while varying configuration. Varying width, head dimension, and parameter count within the same configuration class (Pythia 410M vs. 1.4B) does not recover observability. Switching from regular to deduplicated Pile at matched architecture (Pythia 1.4B vs. 1.4B-deduped) does not recover it. Varying configuration at held training (other Pythia sizes) produces the full range of observability values between 0.21 and 0.38 (Figure 3). Two Pile variants, two sizes with a $3.5\times$ parameter gap, and three 7-seed runs each produce $\rho_{\text{partial}} \approx 0.10$.

Collapse is configuration-specific, not monotone in depth or heads. Collapse is not a monotonic function of any single architectural axis. Deeper models do not collapse harder: 32-layer Pythia (2.8B at 0.208, 6.9B at 0.240) and 36-layer Pythia (12B at 0.238) produce healthy signal. Higher head counts do not collapse harder: 32-head and 40-head configurations are healthy. Depth and head count co-occur at both collapse points (depth=24 AND heads=16) and no other Pythia size matches either value. The two axes are perfectly confounded within this suite.

Baselines and validity. Pythia 70M’s random untrained probe produces 0.078, elevated relative to larger Pythia sizes but still $3.9\times$ below the learned probe’s 0.301. The finding holds with this elevated baseline. The target validity tests from Section 3 apply unchanged here: the probe training protocol is identical, residualizing against the same confidence covariates.

Within-Pythia statistical test. If you were allowed to label any two Pythia configurations as collapsed, the actual (24-layer, 16-head) architecture label produces the largest gap. An exact

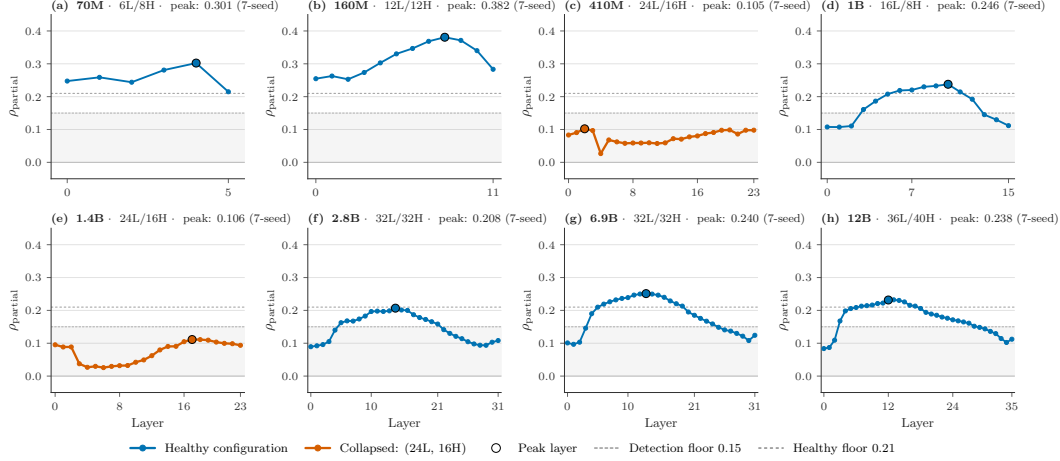


Figure 3: Pythia layer profiles under controlled training. Six healthy configurations peak at mid-to-late depth. Two collapse configurations stay flat across all layers. No layer choice rescues the collapse.

permutation test formalizes this: under the null (arbitrary 2-vs-6 partitions), the observed gap of 0.163 is the unique maximum of the $\binom{8}{2} = 28$ possible partitions ($p = 0.036$). Including the 1.4B-deduped replication as a third collapsed point lowers the permutation p -value to 0.012 ($1/84$). F-test, η^2 , and leave-one-configuration-out robustness appear in [Appendix A](#).

The collapse is signal, not probe failure. A shuffle test at the collapse configuration confirms the residual value is real signal, not probe artifact. At Pythia 1.4B layer 17, probes trained on randomly permuted labels yield $\rho_{\text{partial}} = -0.002 \pm 0.036$ across 10 permutations, while the real probe scores 0.106 and exceeds every shuffled run. The permutation-null Gaussian approximation, code-path sanity check, and GPT-2 cross-comparison appear in [Appendix A](#).

Linear probe is not the bottleneck. A two-layer MLP probe with held-out hyperparameter selection does not close the gap at Pythia 410M or Pythia 1.4B ([Table 5](#)); the same conclusion at Llama 3.2 3B is reported in [Section 3](#). Across the three collapse configurations under two training recipes, the best tested nonlinear probe never crosses the healthy floor of 0.21. The collapse reflects a representational property rather than a limit of linear probing.

Table 5: Swept-hyperparameter nonlinear probe with held-out selection (fit on train, select HP on WikiText validation, report on WikiText test). Grid: hidden $\in \{64, 128\}$, lr $\in \{0.01, 0.001, 0.0001\}$, epochs $\in \{20, 50\}$. Δ is swept-HP MLP minus linear on test. Above-floor configurations (healthy on the main validation protocol) gain $\leq +0.07$ on test. The three collapse configurations, tested under two training recipes (Pythia held-recipe; Llama production) and at two layer choices for Llama 3B, gain $+0.014$ to $+0.043$ on test and remain below the validation healthy floor of 0.21. Test values may fall slightly below validation due to split sampling ([Appendix B](#)); no tested probe recovers healthy-range signal where the linear probe fails.

Configuration	Peak layer	Linear test	Swept-HP MLP test	Δ
<i>Above-floor configurations (healthy on main validation protocol)</i>				
GPT-2 124M	L8	0.293	0.339	0.046
Qwen 2.5 0.5B	L19	0.219	0.254	0.035
Qwen 2.5 1.5B	L19	0.277	0.325	0.048
Qwen 2.5 3B	L25	0.286	0.325	0.038
Qwen 2.5 7B	L17	0.244	0.295	0.051
Qwen 2.5 14B	L30	0.228	0.296	0.067
Gemma 3 1B	L11	0.174	0.198	0.024
<i>Collapsed configurations</i>				
Llama 3.2 3B	L0	0.088	0.102	0.014
Llama 3.2 3B (layer check)	L27	0.104	0.147	0.043
Pythia 410M (24L, 16H)	L2	0.126	0.139	0.014
Pythia 1.4B (24L, 16H)	L17	0.129	0.172	0.043

Deltas computed from unrounded values.

None of the tested alternative explanations accounts for the collapse. Signal could be confidence leakage, but confidence covariates are applied throughout. It could be a linear-readout limitation, but tested nonlinear probes do not cross the healthy floor. It could be layer misselection, but collapsed models show no healthy-range layer at any depth. It could be underpowering, but collapsed models remain collapsed at token budgets well above the budget threshold. It could be architectural incapability, but both configurations form and strengthen the signal early in training. What remains is whether the mid-layer signal is merely redundant with the output representation. That is the test r_{OC} performs.

Output-independence collapses at the same configurations. The output-controlled residual r_{OC} tracks ρ_{partial} across the Pythia suite and collapses at the same three runs (Table 4). Healthy configurations produce r_{OC} between 0.088 (Pythia 2.8B) and 0.169 (Pythia 160M). The three 24-layer, 16-head runs produce 0.043 (410M), 0.003 (1.4B), and -0.012 (1.4B-deduped): near-zero or negative output-independent component at each collapse point. Separation is wider on r_{OC} than on ρ_{partial} because collapsed r_{OC} values reach zero or negative. At the 24-layer, 16-head class, the confidence-controlled signal falls to the detection floor and the output-controlled component falls to near zero. Architecture configuration therefore determines output-independence under controlled training, not just partial correlation. What collapses is not merely a metric but the output-independent component the controls isolate.

The cross-family generalization of this property appears in Section 5: architectures that lose confidence-controlled observability also lose output-independence across the 26 models measured in the paper.

Observability collapse emerges during training. Both matched-width configurations form and strengthen the quality signal early in training. We evaluate two Pythia configurations with matched hidden dimension ($d = 2048$) at 10 checkpoints from step 256 ($\approx 0.5B$ tokens) to step 143,000 ($\approx 300B$ tokens), recomputing losses, confidence covariates, residual targets, probe heads, and output-side readouts at each checkpoint under the same protocol as the final-checkpoint experiments. At the earliest measured checkpoint, both the 1B (16-layer, 8-head, healthy at convergence) and the 1.4B (24-layer, 16-head, collapsed at convergence) show healthy ρ_{partial} (0.315 and 0.297) and positive r_{OC} (0.079 and 0.065). By step 1,000 ($\approx 2B$ tokens), the 1.4B configuration has strengthened further ($\rho_{\text{partial}} = 0.314$, $r_{OC} = 0.113$). The 24-layer, 16-head architecture is capable of representing the output-independent decision-quality signal early in training.

The trajectories diverge during mid-training (Figure 4). Both configurations dip below the detection floor around step 16,000. The 1B configuration recovers by step 32,000 and converges healthy ($\rho_{\text{partial}} = 0.246$, $r_{OC} = 0.120$); the 1.4B configuration does not, converging after a transient recovery at step 64,000 to $\rho_{\text{partial}} = 0.106$ and $r_{OC} = 0.003$, matching the final-checkpoint collapse in Table 4. The erasure is selective: the output-independent fraction ($r_{OC}/\rho_{\text{partial}}$) drops from 36% at step 1,000 to 3% at convergence in the 1.4B, while it grows from 33% to 49% in the 1B (Table 11).

Perplexity declines monotonically in both configurations through the observability dip. The 1B converges to perplexity 20.6 with $\rho_{\text{partial}} = 0.246$; the 1.4B converges to lower perplexity 18.4 with $\rho_{\text{partial}} = 0.106$, so improved language-modeling loss does not imply preserved observability. The healthy configuration achieves both, so preservation does not require trading off predictive performance. Predictive capability and observability are not inherently coupled in either direction.

Architecture does not prevent the signal from appearing. It determines whether training preserves or erases it. The collapse is therefore not static incapacity: the signal forms, strengthens, and is later erased. This is the empirical hinge for signal engineering (Definition 3): an intervention window opens between formation and erasure.

Under controlled training, configuration determines whether the signal survives. Section 5 asks whether independent recipes reproduce the preservation-versus-collapse pattern at different collapse points.

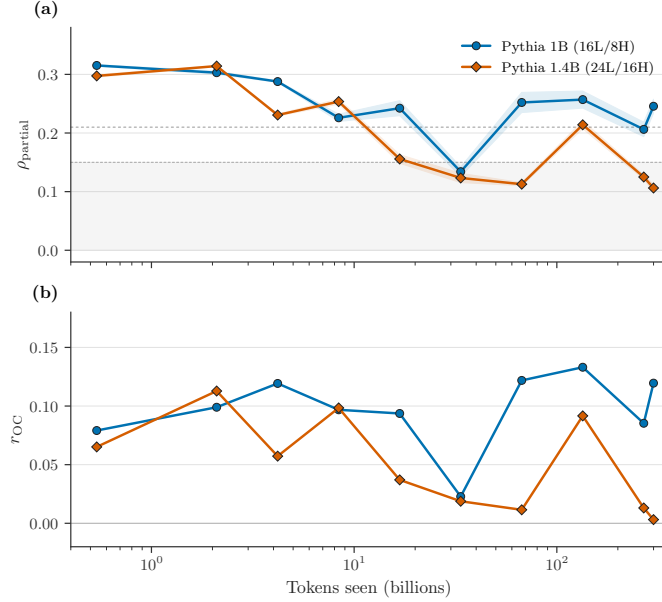


Figure 4: Training improves prediction while erasing observability. Both configurations begin with healthy ρ_{partial} at the earliest measured checkpoint. Both dip during mid-training. The 16-layer, 8-head configuration recovers and converges healthy. The 24-layer, 16-head configuration converges collapsed. Final perplexities are comparable (18–21), so the divergence is monitorability, not predictive capability. Dashed lines in panel (a): detection floor (0.15) and healthy floor (0.21).

5 ACROSS INDEPENDENT TRAINING RECIPES

Section 4 established the controlled within-recipe result: under Pythia, configuration determines whether training preserves observability. This section asks whether the preservation-versus-collapse pattern recurs under independent training recipes, and what ceiling it imposes on downstream monitoring. Under a shared evaluation protocol with token budgets scaled by hidden dimension, six families (Qwen 2.5 (Yang et al., 2024), Llama (Grattafiori et al., 2024), Gemma 3 (Gemma Team, 2025), Mistral (Jiang et al., 2023), Phi-3 (Abdin et al., 2024), and GPT-2) show preserving, marginal, and collapsed regimes. These comparisons are observational; Pythia remains the controlled causal anchor.

Two families preserve observability across scale. Qwen 2.5 preserves observability across a $64\times$ parameter range: ρ_{partial} stays between 0.21 and 0.28 from 0.5B through 32B (Table 6), and the output-controlled residual r_{OC} remains positive at every scale tested. GPT-2 shows the same scale-invariance under a different template: from 124M to 1.5B, ρ_{partial} remains between 0.29 and 0.30 (Table 10) and r_{OC} stays positive at every size (0.084–0.151). Under fixed family templates, scale alone does not change observability under these recipes. Instruction tuning preserves the same pattern across tested Qwen instruct sizes (Appendix I).

At matched 3B scale, Qwen and Llama diverge sharply. The difference is not subtle: Qwen 2.5 retains strong observability (0.263) while Llama sits near the detection floor (0.091), a $2.9\times$ gap. The per-probe-seed distributions do not overlap: every Qwen 2.5 seed (0.225 to 0.288) exceeds every Llama seed (0.084 to 0.102). The gap survives a nonlinear confidence check: under the nonlinear MLP on $[\hat{p}_{\text{max}}, \|\mathbf{h}\|]$ (Section 3), Llama 3B falls to -0.005 .

GPT-2 (124M–1.5B), Mistral 7B (0.313), Phi-3 Mini (0.300), Gemma 3 1B (0.216), and Gemma 3 4B (0.191) confirm signal across four additional families (Figure 5).

Table 6: Observability across six architecture families under shared evaluation protocol. Token budgets scaled by hidden dimension (350 ex/dim, 600 for Qwen 2.5 0.5B). Llama 1B matches high-observability families; 3B and 8B drop to near the detection floor.

Model	Family	Params	Peak layer	$\rho_{\text{partial}}^a \pm \text{std}$	r_{OC}	τ_{seed}	Rand. head
Qwen 2.5 0.5B	Qwen	0.5B	L19 (79%)	0.215 0.020	0.059	0.959	-0.055
Gemma 3 1B	Gemma	1B	L11 (42%)	0.216 0.011	0.161	0.784	-0.004
Llama 3.2 1B	Llama	1.2B	L13 (81%)	0.286 0.006	0.120	0.995	0.012
Qwen 2.5 1.5B	Qwen	1.5B	L18 (64%)	0.275 0.032	0.127	0.953	-0.051
GPT-2 XL	GPT-2	1.5B	L36 (75%)	0.296 0.017	0.084	0.829	0.002
Qwen 2.5 3B	Qwen	3B	L25 (69%)	0.263 0.021	0.144	0.925	0.023
Llama 3.2 3B	Llama	3B	L0 (0%)	0.091 0.006	0.031	0.998	-0.002
Phi-3 Mini	Phi	3.8B	L17 (53%)	0.300 0.002	0.144	0.958	-0.008
Gemma 3 4B	Gemma	4.3B	L6 (18%)	0.191 0.007	0.059	0.802	-0.010
Mistral 7B	Mistral	7B	L22 (69%)	0.313 0.001	0.156	0.995	0.014
Qwen 2.5 7B	Qwen	7B	L17 (61%)	0.255 0.019	0.137	0.964	-0.013
Llama 3.1 8B	Llama	8B	L1 (3%)	0.093 0.012	-0.002	0.994	0.005
Qwen 2.5 14B	Qwen	14B	L30 (62%)	0.214 0.032	0.096	0.851	0.009
Qwen 2.5 32B	Qwen	32B	L44 (69%)	0.237 0.004	0.103	0.871	-0.019

^a Validation split (held-out seeds, $n = 7$ for all models). Across 12 cross-family models with test splits, rankings preserve and mean absolute gap is 4.3% (Appendix B).

Llama 3B/8B peak at L0/L1: no layer exceeds +0.12; reported peak is argmax of a flat profile.

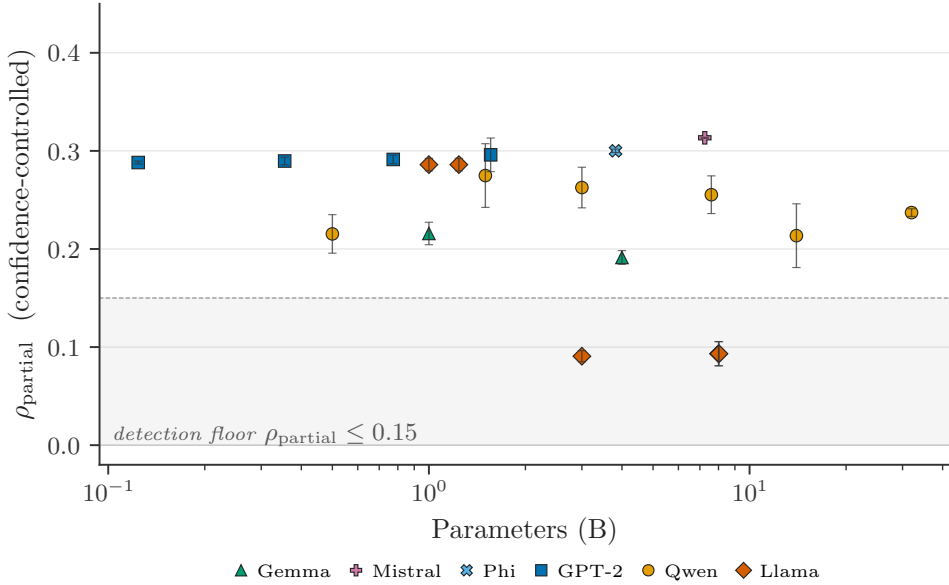


Figure 5: Observability across six families. Llama (red) drops from 0.286 at 1B to 0.091 at 3B. GPT-2 spans 0.29 to 0.30 across the four sizes. Non-Llama families remain above the detection floor. Shaded band: detection floor.

The Llama cliff. The Llama cliff is not gradual (Figure 1, left). Llama 3.2 1B produces healthy observability ($\rho_{\text{partial}} = 0.286$), while Llama 3.2 3B falls near the detection floor (0.091) and Llama 3.1 8B confirms the collapsed regime (0.093). The 1B-to-3B transition changes both architecture (16-layer, 32-head, 2048-hidden vs. 28-layer, 24-head, 3072-hidden) and distillation provenance (Meta, 2024), and the 3B-to-8B extension crosses the 3.2/3.1 recipe boundary. The Llama comparison does not isolate a single causal axis; Pythia supplies that controlled test. It shows that the within-family discontinuity recurs under a different recipe, at a different configuration from Pythia (Llama: 28-layer, 24-head; Pythia: 24-layer, 16-head).

Training recipe shifts which configurations collapse. Mistral 7B v0.3 and Llama 3.1 8B share the same high-level 32-layer, 32-head, 4096-hidden shape yet differ in observability (0.313 vs. 0.093). Within a recipe, configuration determines preservation; across recipes, recipe choice shifts which configurations collapse. No tested configuration collapses universally across recipes. Across recipes,

the architecture-recipe choice bounds the upstream ceiling on observability. Signal engineering (Definition 3) makes recipe a design choice alongside architecture.

Output-independence tracks the cross-family pattern. The output-controlled residual extends the within-Pythia result across all measured families: models that lose confidence-controlled observability also lose the output-independent component measured by r_{OC} (Figure 6). Preserving configurations across all measured families fall on a linear trend; five collapsed runs (Llama 3B, 8B; Pythia 410M, 1.4B, 1.4B-deduped) sit near the origin on both axes.

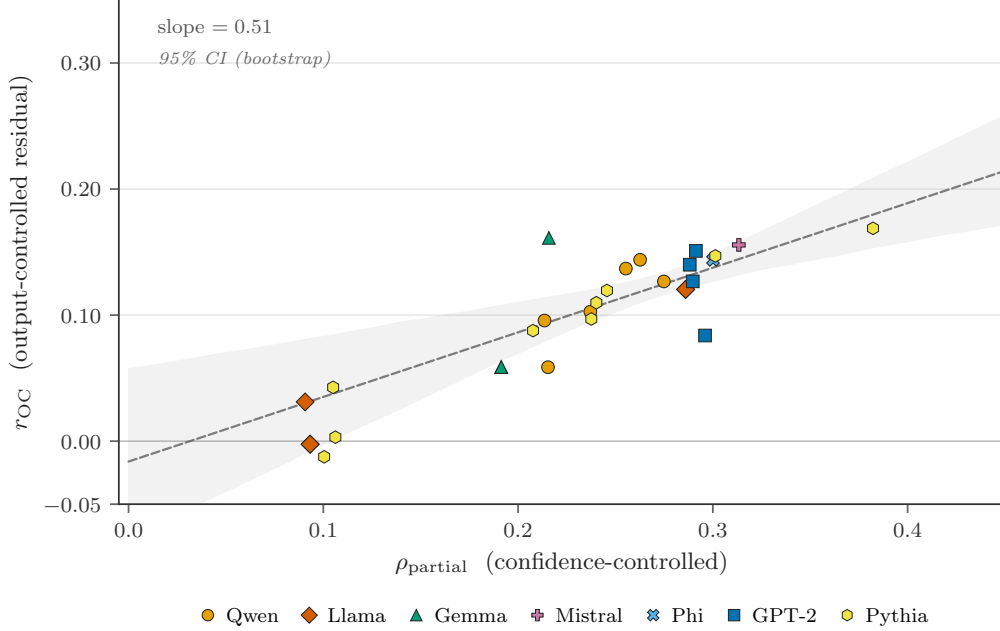


Figure 6: Output-independence (against the tested final-layer readout) collapses with observability. Output-controlled residual (r_{OC}) vs confidence-controlled partial correlation (ρ_{partial}) across 26 models (17 cross-family plus 9 Pythia). Healthy configurations fall on a linear trend (dashed line, slope 0.51). Five collapsed runs (Llama 3B, Llama 8B, and three Pythia replications at the 24-layer, 16-head class) fall near the origin on both axes and are excluded from the healthy-configuration fit. Architecture-dependent observability and output-independence are two views of the same collapse phenomenon.

Statistical scope. The matched-scale 3B pair is the main cross-family comparison: under identical protocol, evaluation data, and token budget per hidden dimension, Qwen 2.5 3B and Llama 3B have non-overlapping probe-seed distributions. Broader cross-family statistics are sensitivity checks rather than the inferential anchor, because four families contribute a single model. A permutation test on model-mean ρ_{partial} ($p = 0.053$) and a mixed-effects model (Equation (6)) both show family-level structure dominating scale; full leave-one-family-out, Jonckheere-Terpstra, analysis-of-covariance, and variance-decomposition analyses appear in Appendix A. The controlled causal claim remains the Pythia suite; the cross-family analyses show that preservation-versus-collapse recurs under independent recipes.

Operational catch is bounded by error structure. Architecture-dependent observability sets an upstream ceiling, but operational catch is also bounded by the error structure. A monitor can only catch errors that are both present in the task and separable from confidence by the observer score. Table 7 therefore reports exclusive catch across both preserving and collapsed configurations at fixed flag rates. At 10% flag rate, the observer catches confidence-invisible language-modeling errors under this loss-based definition. For language modeling, an error is a token with loss above the median, so errors occupy half of all tokens. Exclusive catch is the fraction of these errors flagged by the observer but not by confidence.

At low flag rates, higher observability tends to yield larger exclusive catch. At 20%, the models converge to a 12–15% band, indicating a ceiling set by the structure of the error set rather than by observability alone. Exclusive catch has a nonzero random baseline because two rankers with fixed

Table 7: Exclusive error catches (observer finds, confidence misses) as a percentage of all errors, across four architecture families at four flag rates. The catch rate tends to be larger for higher- ρ_{partial} models at low flag rates but converges to 12–15% at 20%, indicating a ceiling set by the error structure rather than by observability.

Model	Family	ρ_{partial}	5%	10%	20%	30%
Mistral 7B	Mistral	0.313	7.3%	11.4%	14.5%	12.6%
GPT-2 124M	GPT-2	0.288	6.9%	10.8%	14.9%	16.5%
Qwen 2.5 7B	Qwen	0.255	6.1%	9.8%	13.0%	12.7%
Qwen 2.5 14B	Qwen	0.214	6.3%	10.4%	13.2%	12.8%
Llama 3.2 3B	Llama	0.091	3.8%	8.4%	11.6%	12.0%
Llama 3.1 8B	Llama	0.093	5.9%	9.7%	12.0%	11.4%

Catch-rate columns are 3-seed means; ρ_{partial} is the 7-seed validation mean from Table 6. Flag rate is the fraction of tokens flagged by each monitor.

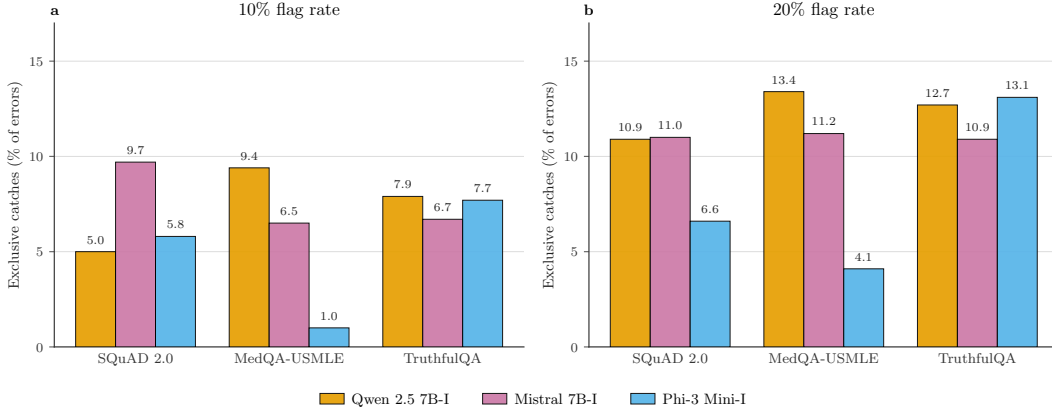


Figure 7: Exclusive catch rates across three tasks and three production instruct models. At 20% flag rate (right), seven of nine cells land in the 10.9–13.4% band, close to the language-modeling ceiling (Table 7). Phi-3 Mini on SQuAD and MedQA are the two exceptions, task-model interactions rather than method floors. All downstream runs use each model’s WikiText-trained probe applied without task-specific training. Task-specific random-ranker baselines are not computed (see Appendix H); rates reflect non-overlapping catches relative to confidence, not excess over a downstream-specific null.

flag rates will disagree by construction. On GPT-2 124M, an analytical random ranker independent of confidence catches 8.1% at 10% and 13.0% at 20%. The observer exceeds random by 2.7 percentage points at 10% and 1.9 at 20%. The margin narrows as flag rate rises.

Downstream transfer near the language-modeling ceiling. The WikiText observer transfers across three question answering (QA) tasks (SQuAD, MedQA, TruthfulQA), catching 10.9–13.4% of confidence-invisible errors (errors flagged by the observer but missed by output-confidence monitoring) at 20% flag rate in seven of nine model-task cells (Appendix H). On the fluent-falsehood subset of TruthfulQA the observer is near chance (area under the ROC curve, AUC 0.499, 0.568, 0.556 across three production instruct models), marking a method boundary supported by consistency across three models: token-level decision-quality observability does not reach locally plausible semantic falsehoods. The WikiText-trained probe also transfers to C4 web text across five families (Appendix G); the trained observer generalizes across domains, while direct C4 training often fails, indicating that clean calibration data matters. These rates establish non-overlapping coverage relative to confidence rather than a demonstrated margin over task-specific random rankers; the language-modeling random-ranker analog is reported above, with task-specific computation deferred to future work.

Across recipes, architecture and training recipe bound the upstream ceiling on activation monitorability: frozen monitors can only exploit the signals training preserved.

6 RELATED WORK

Prior work on activation probes, hidden-state monitors, output-side uncertainty, and monitorability usually begins with a trained model and asks what signal can be extracted from its hidden states, generated text, or output probabilities. This paper asks an upstream question: whether architecture and training preserved the signal a monitor would need. Decision quality is the first target; the broader object is target-specific observability, where the question is which safety- or reliability-relevant signals remain internally readable after training. This framing connects probing validity, behavioral activation directions, output-side uncertainty, deployment-time monitoring, and model-design work on architecture and training dynamics.

Probing validity and output-independent signals. Linear probes were introduced by [Alain and Bengio \(2017\)](#) to monitor intermediate representations. Probing has a known gap between what a probe finds and what the model uses ([Belinkov, 2022](#)). [Hewitt and Liang \(2019\)](#) address one side with control tasks: selectivity measures whether a probe finds more than it could from random linguistic labels on the same representation. A probe can pass this test and still be redundant with the model’s own output confidence. Partial correlation with confidence covariates reframes the probing-validity question from “more than random?” to “more than the output already provides?” On GPT-2 124M, raw Spearman is 0.514, selective but half confidence-redundant (0.289 after covariates). Across 14 models with control-sensitivity data, controlling for confidence removes $60.3\% \pm 7.0\%$ of the raw probe signal, ranging 43.8–75.8% across the cohort ([Section 3](#)). Hidden states encode answer correctness before generation completes ([Zhang et al., 2025](#)), and chain-of-thought success is predictable from pre-completion representations ([Afzal et al., 2025](#)); both works evaluate discrimination between internal representations and correctness labels without controlling for output confidence, so the output-independent component of their reported signals is unmeasured. Output-side readouts sharpen the same distinction: r_{OC} asks whether the surviving signal remains after a trained final-layer predictor has been given access to the output representation, so the comparison is not merely probe versus confidence but mid-layer monitor signal versus tested output-side readout. Our scope also extends to variation that fixed-model studies do not surface: the confidence-controlled signal survives on six families and collapses on Llama 3B and 8B under the same protocol, and checkpoint dynamics show the collapse is training-emergent rather than a static property of the converged model.

Truthfulness, factuality, hallucination, and behavioral directions. Linear directions for truthfulness have been found via unsupervised consistency ([Burns et al., 2023](#)), geometric structure ([Marks and Tegmark, 2024](#)), cross-model universality ([Bürger et al., 2024](#)), and inference-time causal intervention ([Li et al., 2023](#)). [Zou et al. \(2023\)](#) read linear directions for behavioral concepts from intermediate layers. [Han et al. \(2025\)](#) train lightweight factuality probes on hidden states for long-form generation, achieving competitive hallucination detection with a single forward pass. [Azaria and Mitchell \(2023\)](#) train classifiers on activations to detect model-generated falsehoods. [Goldowsky-Dill et al. \(2025\)](#) use linear probes to detect strategic deception from activations, finding high AUC (0.96–0.999) but noting the probe sometimes fires for the topic of deception rather than genuine intent. These methods target semantic or behavioral labels and evaluate by discrimination metrics. Our observer targets a different object (confidence-residual per-token loss) and measures a different property (architecture-dependent preservation vs. collapse); none of these works test cross-architecture variation or training-emergent collapse.

Output-side uncertainty, calibration, and selective prediction. Output-level methods operate on what the model has already exposed. Calibrated-confidence ranking ([Guo et al., 2017](#)) captures most of the operationally useful signal; the 8–12% exclusive error rate at 10% flag rate is the residual it leaves uncovered. Scalar monotone recalibrations preserve Spearman ranks; a nonlinear MLP on $[\hat{p}_{\max}, \|\mathbf{h}\|]$ absorbs no additional signal ($\rho_{\text{partial}} = 0.319$ vs. 0.289), so nonlinear miscalibration is not the residual’s source. Language models can estimate their own accuracy in some settings ([Kadavath et al., 2022](#)), but the output-side predictor on the full last-layer representation still leaves $r_{OC} > 0$ at every tested GPT-2 scale. Verbal uncertainty is governed by a single linear feature in representation space ([Ji et al., 2025](#)), correlating only moderately with semantic uncertainty; verbal assertiveness correlates with output confidence, so this feature may overlap with what our covariates remove. [Kuhn et al. \(2023\)](#) cluster sampled generations by meaning across multiple forward passes; [Kossen et al. \(2025\)](#) approximate semantic entropy from hidden states via linear probes; both predict an output-level quantity. Output-level retrieval-augmented generation (RAG) detectors also operate

on generated text: NLI faithfulness (Honovich et al., 2022), sample consistency (Manakul et al., 2023), and atomic-claim factual precision (Min et al., 2023) all target what the model already exposed. Selective prediction methods including early exit (Schuster et al., 2022) and verbalized uncertainty for LLMs (Wen et al., 2025) make exit decisions on output-level signals. The observer catches a non-overlapping error class: tokens where the model is confident and the output gives no warning but the mid-layer activations encode the failure. On SQuAD 2.0, MedQA-USMLE, and TruthfulQA, a WikiText-trained probe catches 10.9–13.4% of errors exclusively at 20% flag rate in seven of nine model-task cells. Output-level and observer methods are complementary because they ask different questions: what the output reveals versus what the activations preserve.

Activation monitoring at scale, monitorability, and evasion. Production probes are fragile across distributions (Kramár et al., 2026). Kramár et al. build production-ready activation probes for a model whose representation encodes the relevant signal; this paper identifies the upstream condition: whether the architecture-recipe choice preserved that signal in the first place. Rule-based activation monitoring (Rozenfeld et al., 2026), dynamic safety monitoring (Oldfield et al., 2026), and monitorability metrics (Guan et al., 2025) measure monitoring quality across deployed models without isolating which upstream choices determine signal preservation. Mechanistic interpretability findings require causal methods and separation of correlation from causation (Sharkey et al., 2025); McGuinness et al. (2025) show that models can learn to evade activation monitors, and Ji-An et al. (2025) find that LLMs have emergent capacity to monitor and modulate their own activation patterns. The observer direction is near-orthogonal to dominant variance on GPT-2 124M (PC1 cosine 0.002), suggesting that decision-quality signals can occupy low-variance subspaces that variance-dominant decompositions and some evasion analyses may miss.

Architecture, scaling, and training as signal-preservation axes. Model-design work usually treats architecture and training recipe as controls for loss, compute efficiency, scale, latency, and downstream capability. Controlled training suites such as Pythia (Biderman et al., 2023) make it possible to ask how these choices affect training dynamics rather than only final task scores. This work adds a different dependent variable to that design space: whether training preserves monitorable internal signals. In Pythia, a decision-quality signal appears early in both matched-width configurations, but one configuration preserves it while another erases it as perplexity improves. Across production families, recipes shift which configurations collapse. Observability therefore connects activation monitoring to architecture search and signal engineering (Definition 3): the question is not only how to read a frozen representation, but how to design training so that oversight-relevant signals are preserved or amplified as readable from frozen activations.

Chain-of-thought monitorability preservation (Korbak et al., 2025; Xiong et al., 2026) addresses a parallel axis on a different signal: the legibility of reasoning traces under training pressure. The two axes are independent: a model could have monitorable chain-of-thought and erased activation signals, or vice versa. This paper’s contribution is the activation-level question with controlled-recipe evidence.

Across these literatures, the common assumption is that the trained model is fixed and the problem is to extract, calibrate, harden, or compare whatever signal remains. Our results move the question upstream. For decision quality, and potentially for other safety and reliability targets, the relevant signal may be formed during training and later preserved, compressed, scattered, or erased. Observability therefore turns activation monitoring into a model-design problem: architecture and training determine what oversight can see, and signal preservation becomes a training-time engineering target.

7 DISCUSSION

Training shapes monitorability. Training does not merely determine what a model predicts. It determines which internal evidence remains available to oversight. In Pythia’s controlled training, both matched-width configurations form the decision-quality signal early, but training preserves it in one configuration and erases it in another while perplexity improves. Predictive optimization and observability outcomes are separable consequences of architecture and training. Capability and monitorability are not inherently in tension: the healthy 16-layer, 8-head Pythia configuration preserves both through convergence, while the 24-layer, 16-head class loses observability. Across

independent recipes the collapse point changes: Llama collapses at a different configuration (1B healthy, 3B and 8B collapsed), and Mistral 7B v0.3 and Llama 3.1 8B share the same architecture (32-layer, 32-head, 4096-hidden) yet differ in observability (Section 5). Across these recipes, loss does not predict observability.

Monitoring ceilings are set at training time. Activation monitoring treats the trained model as fixed, but the model may already lack the signal. In collapsed configurations, full layer sweeps, swept nonlinear probes, token-budget checks, and tested output-side readouts all fail to lift observability above the detection floor. The dynamics show why: training actively erases a signal the architecture initially formed. Once architecture and training are fixed, monitors are bounded by the observability of the trained model. Scale and probe-width sweeps confirm the bound (Table 6, Section 3). Frozen monitors read signals that training preserves or amplifies.

Observability as a design axis. Observability has two operational uses. The immediate use is model selection: a lab evaluating trained candidates can measure observability before deployment, alongside loss and capability. The deeper use is design. Architecture and training-recipe choices jointly bound whether monitorable signals are preserved, amplified, compressed, or erased through training. This is an upstream property of the trained model, and it bounds what any frozen monitor can read.

The downstream catch results show why the design axis matters (Figure 7). At 20% flag rate, seven of nine model-task cells across SQuAD, MedQA, and TruthfulQA fall in the 10.9–13.4% exclusive-catch band, with the observer catching about one in seven of Qwen 2.5 7B Instruct’s MedQA errors that confidence does not flag. These results show the operational stakes of preserving or amplifying signal at training time.

From monitoring to signal engineering. Once a model is frozen at a collapsed configuration, no tested probe or output-side readout recovers the signal. Checkpoint dynamics show the signal was not impossible to represent in that architecture: it appeared early and was later erased. The architecture had the capacity, and training did not preserve it. This opens an intervention window between formation and erasure, and makes monitorability a design problem, not only a diagnostic one. Signal engineering (Definition 3) names this design problem. Decision quality is the first demonstrated target. The paper establishes the axis, not a signal engineering training intervention.

Signal engineering has a geometric implication. On GPT-2 124M, the observer direction is nearly orthogonal to dominant representation variance (PC1 cosine 0.002), and reconstruction-trained SAEs partially obscure it. Whether the signal lands in readable geometry is itself a training outcome.

Observability beyond decision quality. Decision quality is one instance of a protocol that generalizes. For any target with specifiable labels and output-side readouts, the same evidence hierarchy asks whether training preserves a readable signal beyond output confidence. Changing the target changes the labels and readouts, not the burden of evidence. The same protocol applies (Section 2). Candidate targets include factuality, refusal appropriateness, retrieval faithfulness, reward-model disagreement, tool-use reliability, and controlled deception labels. Whether these targets exhibit the same architecture-recipe dependence or collapse at the same configurations remains open.

Limitations and future work. The paper’s claims are tiered by evidence strength. Each tier names its limitation and the advancing experiment.

Within-recipe causation. Checkpoint dynamics provide the primary within-recipe temporal evidence: both matched-width configurations form the decision-quality signal early, then training erases it in the 24-layer, 16-head class while the 16-layer, 8-head configuration recovers, at comparable final perplexity. Within the Pythia suite, depth and head count co-vary at the (24-layer, 16-head) class, so isolating which axis drives the divergence requires single-axis controlled training. Why one configuration recovers from the mid-training dip while the other does not is open. *Advancing the claim: single-axis controlled training at matched scale, and finer-grained checkpoint analysis during the divergence window.*

Observational replication across recipes. Cross-family evidence and the Llama cliff show the pattern replicates under different training recipes. Cross-recipe causation is the next test, requiring controlled training across recipes. *Advancing the claim: controlled training at matched scale across recipes.*

Operational transfer. Operational utility depends on task-model factors beyond token-level observability. The next bound is direct overlap with single-pass hallucination detectors (Kuhn et al., 2023; Kossen et al., 2025) on the confidence-invisible subset. *Advancing the claim: direct comparison with semantic-entropy variants on the confidence-invisible subset.*

Method scope. A separate residualizer-fit split preserves the observability regime on 4 representative models (Appendix B.2). The tested output-side capacity bottleneck is not the explanation for r_{OC} . Stronger or qualitatively different output-level monitors remain outside scope. Adaptive evasion against the trained observer (McGuinness et al., 2025) is the next threat-model study. This paper measures the upstream signal availability that any monitor depends on. For decision-quality or per-token error targets, results reported without confidence covariates may substantially overstate the output-independent component: in our model suite, controlling for confidence removes 60.3% of raw signal on average across 14 models. *Advancing the claim: tests against stronger output-side readouts and adaptive evasion studies.*

Scale. The largest model tested is Qwen 2.5 32B. Whether observability persists at frontier scale is unknown, although Qwen 2.5 0.5B–32B and GPT-2 124M–1.5B show no degradation across the tested within-family ranges. *Advancing the claim: observability measurements at frontier scale, particularly under training-recipe variations adopted at frontier compute budgets.*

Architecture and training shape the geometry of the residual stream, determining whether the decision-quality signal remains readable or scatters beyond the reach of activation monitors. Activation monitorability is decided upstream, before monitor design begins. Training sets the observability ceiling.

8 CONCLUSION

Architecture decides whether activation monitoring catches the confident errors output-confidence monitoring misses. Training decides what remains readable, before any monitor is built. In Pythia’s controlled training, both matched-width configurations form the decision-quality signal early. Training then preserves it in Pythia 1B (16-layer, 8-head) and erases it in Pythia 1.4B (24-layer, 16-head). Both improve in perplexity, and the 1.4B reaches lower final perplexity than the 1B. Predictive capability and monitorability are not inherently in tension. The output-controlled residual collapses with observability. Across independent recipes, the collapse point shifts but the same split between preservation and collapse recurs.

Monitorability is an upstream design axis, alongside loss and capability. It is measurable on the trained model and shaped through architecture and training-recipe choices that preserve readable decision-quality signals. Decision quality is the first demonstrated target. The approach generalizes to other safety and reliability targets. Monitoring is bounded by the observability of the trained model. The next problem is signal engineering: training models not only to predict well, but to preserve or amplify monitorable signals in activations. Architecture sets the conditions for observability, and training determines what remains readable.

REPRODUCIBILITY AND DISCLOSURE

Reproducibility. Code version v5.1.0 (commit e0b505ed6e7aab76e6b3d510972b16396097cb02) is tagged at <https://github.com/tmcarmichael/nn-observability/tree/v5.1.0>. The codebase and results are archived at Zenodo under the concept DOI [10.5281/zenodo.19435674](https://doi.org/10.5281/zenodo.19435674), which resolves to the latest version (the v5.1.0 snapshot at the time of submission). Per-dataset documentation lives in DATA.md at the code release. Committed results JSONs are the source of truth; each paper-scope result includes a provenance block recording the model revision, generating script, timestamp, and compute device. Hugging Face model revisions are pinned in results/model_revisions.json, and dataset revisions in results/dataset_revisions.json. Every entry in model_revisions.json is programmatically verified against the Hugging Face API; the latest report under results/manifest_verification/ records the verification timestamp and exact-SHA-match status per entry, and is regeneratable via `python scripts/verify_manifest_revisions.py`. The full result-file inventory is published as a Croissant 1.1 metadata descriptor at croissant.json, validated against the official MLCommons spec by just `check-croissant`. From a clean clone, `uv sync -frozen -extra transformer` prepares the environment. *Verification (no GPU,*

seconds). `uv run pytest tests/ -q && just check` validates every committed result JSON against its schema, every manifest entry against the Hugging Face API, every paper-cited macro against its source files and key paths, and every scope against `analysis/load_results.py`. *Full regeneration (single H100/A100, hours-to-days)*. Two recipes drive the bulk: `just pythia-suite` and `just downstream-all`. Cross-family and other experiments have their own script. Software: Python 3.12, PyTorch 2.8, and Hugging Face Transformers, managed with `uv`.

LLM disclosure. Claude (Anthropic) assisted with Python scripts for GPU data collection and statistical analysis, experiment notebooks, prose editing, and section-level consistency checks against the results JSONs. ChatGPT (OpenAI) was used for review and proofreading.

Broader impact. Architecture and training recipe bound whether activation monitoring has a signal to exploit on a deployed model. Monitorability is partially decided at training time, not bolted on later. In high-stakes domains such as medical QA and context-grounded generation, observability is a pre-deployment measurement alongside latency and accuracy. The observer flags a specific class of confidence-invisible failures. It does not detect fluent factual errors, the boundary demonstrated by TruthfulQA, and has not been tested against adaptive attacks. McGuinness et al. (2025) demonstrate that activation monitors can be evaded, and the same threat model applies here: the signal does not ride a dominant variance direction (PC1 cosine 0.002). Deployment should treat activation monitoring as complementary to output-level checks, not as a replacement.

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APPENDIX

A STATISTICAL HARDENING

This appendix and the three that follow supplement [Sections 2](#) and [3](#). Together they cover the methodology hardening package: statistical hardening (this section), layer and split validation ([Appendix B](#)), confidence-control audits ([Appendix C](#)), and signal characterization ([Appendix D](#)).

A.1 20-SEED STATISTICAL HARDENING

20 independent observer heads at layer 11 of GPT-2 124M (seeds 42–61): $\rho_{\text{partial}} = 0.282 \pm 0.001$, 95% CI $[0.2817, 0.2827]$, per-seed range $[0.279, 0.284]$, seed agreement 0.993. Layer 11 is the peak layer under the 20-seed hardening protocol; the main 7-seed cross-family protocol peaks at layer 8 with 0.288, consistent with the ± 0.015 top-3 flatness reported in [Section 3](#). This CI covers probe initialization variance on fixed activations and a fixed dataset. It does not capture uncertainty from dataset, checkpoint, or model family variation.

A.2 QWEN 14B SEED DISTRIBUTION

Qwen 2.5 14B shows a bimodal seed distribution ($\tau_{\text{seed}} = 0.851$): per-seed values cluster near 0.185 (four seeds) or 0.251 (three seeds), and the reported mean (0.214) averages across both clusters. The scaling claim ([Section 5](#)) does not depend on the exact value: every seed lands between 0.18 and 0.25, well above the detection floor and in line with the rest of the Qwen family. The bimodality is preserved under resampling; characterizing the geometric relationship between the two clusters would require per-seed direction data we did not collect.

A.3 SUPPLEMENTARY STATISTICAL TESTS

Variance decomposition. A three-level decomposition attributes 56.8% of the variance to family membership, with three singleton-cell families inflating the between-family share by construction; the matched 3B pair ([Section 5](#)) and the Pythia controlled suite ([Section 4](#)) are the inferential tests.

Uncertainty on η^2 . A bootstrap over model means (10,000 resamples with replacement, $n = 17$) gives a 95% CI of $[0.29, 0.98]$ on η^2 , with the upper bound at the ceiling. The point estimate is close to the maximum because within-family variance is small relative to between-family variance at matched scale; the wide CI reflects the $n = 17$ sample and the singleton-cell structure above, not instability of the measurement.

Leave-one-family-out sensitivity. Removing any single family keeps the permutation test significant ($p \leq 0.125$ in all cases; weakest exclusion: without GPT-2, $p = 0.125$; without Gemma, $p = 0.061$; without Llama, $p = 0.002$). The result is not driven by any single family.

Jonckheere-Terpstra within-Qwen 2.5 trend. Extending the within-Qwen 2.5 sweep to 32B yields a non-monotonic six-point trajectory (0.215, 0.275, 0.263, 0.255, 0.214, 0.237). The Jonckheere-Terpstra test does not support a monotonic direction ($p = 0.210$; 55.1% of cross-group comparisons decline, near the 50% chance baseline), and ρ_{partial} is bounded between 0.21 and 0.28 across $64\times$ scale. The cross-family separation in [Section 5](#) carries the inferential weight; the within-family scaling story is null. The mixed-effects scale coefficient ($p = 0.285$) confirms within-family variation does not explain cross-family differences.

Supplementary ANCOVA. ANCOVA with scale and family yields family $F(5, 112) = 29.4$, $p = 5.8 \times 10^{-19}$; $\log_{\text{params}} F(1, 112) = 11.06$, $p = 0.00$. Per-seed observations are not independent (shared data and layer selection), so these p -values are anticonservative. The mixed-effects model is the primary test because its random intercepts per model absorb within-model correlation from shared data and layer selection.

Within-Pythia statistical tests. The 8-configuration permutation test ([Section 4](#)) is exact: $\binom{8}{2} = 28$ possible 2-vs-6 partitions, and the (24-layer, 16-head) labeling is the unique partition producing the largest healthy-minus-collapsed gap in ρ_{partial} (0.163), so $p = 0.036$. The healthy mean is 0.269 ($n = 6$, range 0.208 to 0.382); the collapsed mean is 0.106 ($n = 2$, range 0.105 to 0.106). Two supplementary checks: a Mann-Whitney U between the two groups yields $U = 0$, exact one-sided $p = 0.036$; leave-one-configuration-out across the 8 sizes preserves $\max(\text{collapsed}) < \min(\text{healthy})$.

on every removal. The standing gap 0.101 is set by Pythia 2.8B (lowest healthy) against 1.4B (highest collapsed); LOO removals only widen it. Treating 1.4B-deduped as a third replication at the collapse configuration, the 9-point 3-vs-6 exact permutation p-value is 0.012 (1/84); the three replications range from 0.100 to 0.106 (width 0.006) despite $3.5\times$ variation in parameters and two Pile corpora.

Selective prediction. Downstream exclusive catch rates for the observer (WikiText-trained, applied without task-specific training) on SQuAD 2.0, MedQA-USMLE, and TruthfulQA are reported in Table 13. Confidence alone has higher single-signal precision at every tested flag rate; the observer’s value is the non-overlapping catch (Section 5).

A.4 EQUIVALENCE TEST (TOST) PROCEDURE

The matched-hyperparameter MLP-vs-linear comparison (Section 3) uses two one-sided tests (Lakens, 2017) under Schuirmann’s intersection-union framing. Per model i , the seed-averaged paired delta $\bar{\Delta}_i$ is the difference in ρ_{partial} between the MLP probe and the linear probe. Across $n = 10$ models, let $\bar{\Delta}$ denote the pooled mean and $\text{SE}(\bar{\Delta}) = s_{\Delta}/\sqrt{n}$ the standard error, with s_{Δ} the unbiased sample standard deviation (ddof = 1). Two one-sample t -statistics with $n - 1$ degrees of freedom test

$$\begin{aligned} H_0^{\text{lower}} : \mu_{\Delta} &\leq -0.03 \quad \text{vs.} \quad \mu_{\Delta} > -0.03, \\ H_0^{\text{upper}} : \mu_{\Delta} &\geq +0.03 \quad \text{vs.} \quad \mu_{\Delta} < +0.03. \end{aligned}$$

Equivalence at family-wise $\alpha = 0.05$ is declared when $\max(p_{\text{lower}}, p_{\text{upper}}) < 0.05$. This maximum is the reported TOST p -value. Equivalently, the 90% CI on μ_{Δ} lies inside ± 0.03 . The margin is set to twice the empirical top-three layer-flatness tolerance ($0.03 = 2 \cdot 0.015$).

B LAYER AND SPLIT VALIDATION

B.1 LAYER SELECTION AND TEST-SPLIT CONFIRMATION

Layer selection is two-stage. A seed-42 sweep produces a candidate set: the top four layers by seed-42 ρ_{partial} , depth-filtered to $\leq 80\%$ of total depth, plus the seed-42 argmax. A 7-seed evaluation (seeds 43–49) at each candidate selects the reported layer by argmax of the 7-seed mean; reported ρ_{partial} is the 7-seed distribution at that layer. Selection over the small candidate set introduces bounded optimism: layer profiles are smooth across the top 40–60% of depth, and across tested models the top three candidate layers differ by less than 0.015 in ρ_{partial} : Mistral 7B top-3 range 0.0006, Phi-3 Mini 0.0009, Qwen 2.5 14B 0.0005, Qwen 2.5 3B 0.004, Llama 1B 0.004, GPT-2 124M 0.009. Selecting any of the top three layers in place of the reported peak would change no model’s ρ_{partial} by more than 0.015 and no cross-family comparison by more than 0.01.

A separate three-seed test split evaluates the same layer on non-overlapping probe initializations. Across 12 cross-family models with test splits, the mean absolute test-vs-validation gap is 4.3% (max 14.1%; Qwen 2.5 1.5B). Nine of 12 models show test < validation; the remaining 3 show test > validation by at most 0.006 absolute. The largest gap (Qwen 2.5 1.5B: 0.236 test vs. 0.275 validation, -14.1%) is consistent with the smaller sample ($n = 3$ vs. $n = 7$): subsampling the validation seeds into groups of three reproduces comparable variance. Cross-family rankings are identical on test data (Qwen 2.5 3B 0.255 vs. Llama 3B 0.091, a $2.6\times$ gap), and within-family flatness is preserved (Qwen 2.5 test means span 0.200 to 0.255 across 0.5B–32B; Qwen 2.5 14B has no test split).

B.2 SEPARATE RESIDUALIZER-FIT SPLIT

The standard protocol fits the OLS residualizer on the same documents used to train the probe. To test whether this in-sample fit creates an artifact in the residual target, we construct a disjoint residualizer-fit pool R and apply R-fit coefficients without refit to the probe-training pool T and the evaluation pool V. The three pools draw disjoint WikiText-103 documents at the same hidden-dim token budget as the canonical protocol; the residualizer is a three-coefficient OLS fit (softmax, activation norm, intercept), and the probe is the same linear head trained on the residualized binary target on T. Per-model diagnostics including per-seed values, β coefficients, token counts, and pos-label fractions appear in the *_residualizer-split.json files in results/.

We run this on 4 models: GPT-2 124M (healthy anchor), Pythia 1B (healthy controlled), Pythia 1.4B (controlled collapse), Llama 3.2 3B (observational collapse). The pre-specified pass criterion is

regime preservation: healthy models satisfy $\rho_{\text{partial}} \geq 0.21$, collapsed models $\rho_{\text{partial}} \leq 0.15$, with ± 0.02 tolerance (Table 8).

Table 8: Residualizer-fit split robustness. Reference ρ_{partial} is the canonical 7-seed mean. Split-fit ρ_{partial} uses an OLS residualizer fit on a disjoint document pool. All four models satisfy the pre-specified criteria. Max direct-baseline $|\Delta\rho_{\text{partial}}| = 0.017$ across the four models; the canonical-reference difference for Pythia 1.4B is 0.033 because the direct collection path gives 0.127 rather than the all-layer-sweep reference 0.106 (both below the detection floor).¹

Model	Regime	Reference ρ_{partial}	Split-fit ρ_{partial}	Criterion	Result
GPT-2 124M	healthy anchor	0.288	0.288	≥ 0.21	pass
Pythia 1B	healthy controlled	0.246	0.263	≥ 0.21	pass
Pythia 1.4B	controlled collapse	0.106	0.139	≤ 0.15	pass
Llama 3.2 3B	observational collapse	0.091	0.093	≤ 0.15	pass

The matched within-Pythia pair is the strongest cell. Pythia 1B (16-layer, 8-head) and Pythia 1.4B (24-layer, 16-head) share recipe, training data, hidden dimension, and probe-token budget; the only difference is the configuration tuple. Under split-fit, the distinction holds: 1B remains healthy at 0.263, 1.4B remains collapsed at 0.139. This matched pair makes an in-sample residualizer-fit artifact unlikely: the same split-fit protocol preserves the 1B/1.4B separation under disjoint residualizer-fit pools.

The OLS coefficients fit on T versus R are nearly identical across all four models. The softmax coefficient varies by at most 0.10 from its base magnitude (the four base values cluster near -5); the activation-norm coefficient and the intercept are essentially unchanged. Positive-label fractions on T differ between β_T and β_R by less than 0.007 absolute in every model. The residualizer is a deterministic function of token statistics, not a knob whose fitting pool meaningfully changes the target distribution.

C CONFIDENCE-CONTROL AUDITS

C.1 CONTROL SENSITIVITY

On GPT-2 124M, raw Spearman of 0.514 drops to 0.289 after softmax-only control, holds at 0.289 after adding norm, and reaches 0.319 under a nonlinear MLP control on $[\hat{p}_{\text{max}}, \|\mathbf{h}\|]$. Softmax is the main confound; norm adds nothing beyond softmax; the residual survives nonlinear deconfounding. Adding logit entropy as a third covariate absorbs 27% on top of the standard covariates (0.211), indicating the observer partially reads the shape of the output distribution, not just the peak.

Table 9: Control sensitivity across three architectures. Softmax dominates absorption in all three; standard controls remove between 39% (Pythia 410M) and 63% (Qwen 2.5 7B) of the raw signal; norm-only does not remove the residual; nonlinear controls leave similar or slightly higher residuals than standard controls; entropy removes additional output-shape information.

Control	GPT-2 124M	Qwen 2.5 7B base	Pythia 410M (collapse)
None (raw Spearman)	0.514	0.650	0.175
Norm only	0.514	0.646	0.212
Softmax only	0.289	0.236	0.092
Standard (softmax + norm)	0.289	0.240	0.107
Nonlinear MLP	0.319	0.259	0.120
Standard + logit entropy	0.211	0.103	0.035

The control ordering is consistent across architectures. Table 9 extends the GPT-2 124M analysis to a healthy cross-family model (Qwen 2.5 7B base) and to a within-Pythia collapse point (Pythia 410M). In all three, softmax dominates absorption and standard (softmax + norm) covariates remove between 39% and 63% of the raw signal. Norm-only absorbs almost nothing on top, and the nonlinear MLP

¹For Pythia 1.4B, the same-pool baseline recomputed under the direct-collection path is 0.127 versus the canonical all-layer-sweep reference value 0.106. Both lie below the detection floor. The $|\Delta\rho_{\text{partial}}|$ statistic uses the direct baseline so that the residualizer split is isolated from the collection path.

control yields a similar residual to the linear standard control. The collapse point shows the same control ordering at a shifted absolute scale.

C.2 VERIFYING “60.3% IS CONFIDENCE”

The fraction of raw signal removed by controlling for confidence covariates varies across families but is consistently large. Across 14 models in 6 families, standard covariates (max-softmax + activation norm) absorb $60.3\% \pm 7.0\%$ of the raw Spearman correlation, ranging 43.8–75.8% across the cohort. The pattern is consistent: confidence is the dominant confound on every architecture tested. The “half to three-quarters” summary in the main text reflects this range.

Absorption gradient tracks observability. Absorbed fraction tracks architecture-dependent observability. Healthy models span 43.8%–64.5%, with GPT-2 124M at the low end and Qwen 2.5 32B at the top. Both Llama collapse points concentrate at the upper end: Llama 3.2 3B at 63.8% and Llama 3.1 8B at 75.8%, the highest in the cohort: what little raw correlation exists there is almost entirely confidence-redundant. The direction of the gradient, higher absorption where the confidence-independent signal is smaller, is the operational reading of the unified claim: architecture-dependent observability and output-independence are the same property.

C.3 TARGET CIRCULARITY

We construct the binary target by regressing per-token loss on $[\hat{p}_{\max}, \|\mathbf{h}\|]$ via OLS and thresholding the residual at zero. [Section 3](#) presents the complete six-test validity argument; we expand on four of those tests here:

1. Hand-designed readouts of the same activations fail to predict the same target ([Section 3](#)), so the target is not trivially recoverable from activation geometry.
2. The probe succeeds on some architectures (Qwen 2.5 3B 0.263) and fails on others (Llama 3B 0.091) under the identical target definition. The variation is in the representations, not the target formula.
3. On Qwen 2.5 7B, within-domain C4 training fails (-0.029 ; Llama 8B also fails at -0.062), so the target does not manufacture signal on all activation geometries even within architectures that succeed on WikiText.
4. Seed agreement on the residualized (confidence-removed) observer scores remains 0.991, so the cross-seed consensus is on the confidence-independent component rather than on a shared confidence pathway.

D SIGNAL CHARACTERIZATION

D.1 SIGNAL COMPOSITION (SHAPLEY DECOMPOSITION)

Four named controls account for about 67% of the raw signal under a Shapley value decomposition, computed across all 24 orderings of the 4 controls. The 4-control Shapley total is not directly comparable to the 2-control (max-softmax + activation norm) absorption range reported in [Section 3](#) because the control sets differ. The controls are: (i) max-softmax confidence, (ii) logit entropy, (iii) geometric typicality (Mahalanobis distance of the activation from the training-set activation mean under the training covariance), and (iv) token frequency (log unigram frequency computed on the probe training corpus). The remaining 33% resists all tested controls. Individual component attributions vary by ordering (confidence ranges from -2% to 26% , entropy from 5% to 36%) because the covariates are correlated, but the total explained fraction and the unexplained residual are stable. Input-side controls (bigram surprisal) do not absorb the signal.

D.2 SIGNAL GEOMETRY

The observer direction is nearly orthogonal to the dominant variance (PC1 cosine 0.002, top 10 PCs capture 3.7%). The signal reads from the low-variance subspace of the residual stream. Standard dimensionality reduction would miss it. As one example, a 24,576-feature reconstruction-trained sparse autoencoder ([Bricken et al., 2023](#)) probe (0.255) underperforms the raw linear observer (0.288)

Table 10: GPT-2 scaling under matched 350 ex/dim, 7-seed protocol. Partial correlation is stable across $12\times$ scale (0.29–0.30). The output-controlled residual r_{OC} is positive at every size (0.084–0.151), so the confidence-residual signal is not fully recovered by the tested final-layer predictor at any scale. The output-controlled fraction $r_{\text{OC}}/\rho_{\text{partial}}$ varies between 28% and 52% without a monotonic trend in size; predictor-capacity sensitivity is addressed cross-family by the Qwen 7B output-side width sweep (Section 3).

Model	Params	Peak layer	ρ_{partial}	$\pm$ std	r_{OC}	$r_{\text{OC}}/\rho_{\text{partial}}$	τ_{seed}
GPT-2	124M	L8 (67%)	0.288	0.001	0.140	49%	0.952
GPT-2 Medium	355M	L17 (71%)	0.290	0.004	0.127	44%	0.940
GPT-2 Large	774M	L25 (69%)	0.291	0.004	0.151	52%	0.943
GPT-2 XL	1558M	L36 (75%)	0.296	0.017	0.084	28%	0.829

despite $32\times$ more input dimensions, with rank correlation 0.70 between the two. A sparse decomposition optimized for reconstruction concentrates on high-variance directions; the observer direction lies in a low-variance subspace that variance-dominant and reconstruction-trained decompositions partially obscure.

D.3 GPT-2 WITHIN-FAMILY SCALING

Across the four GPT-2 sizes (124M, 355M, 774M, 1.5B), partial correlation lands in 0.29–0.30 and the output-controlled residual in 0.084–0.151, with output-controlled fraction $r_{\text{OC}}/\rho_{\text{partial}}$ in 28%–52% (Table 10). All four are evaluated under the canonical 350 ex/dim, 7-seed protocol used throughout the cross-family suite. The within-family scaling test pairs with the Qwen 2.5 0.5B–32B sweep (Section 5) under a different architecture template.

E CHECKPOINT DYNAMICS

Table 11 reports the full checkpoint trajectory for the two matched-width Pythia configurations ($d = 2048$) analyzed in Section 4. Both configurations produce healthy ρ_{partial} and positive r_{OC} at step 256 ($\approx 0.5\text{B}$ tokens). The 1.4B (24L/16H) configuration peaks at step 1,000 ($\rho_{\text{partial}} = 0.314$, $r_{\text{OC}} = 0.113$), then declines. The 1B (16L/8H) configuration follows a similar early trajectory, dips below the detection floor at step 16,000, and recovers by step 32,000. Both configurations converge to comparable perplexity (18–21) at the final checkpoint. Seed agreement dips and recovers in both configurations; the 1.4B ends at $\tau_{\text{seed}} = 0.947$ alongside collapsed $\rho_{\text{partial}} = 0.106$. Peak layer trajectories differ between configurations. The 1B starts at L12 (steps 256–1k), shifts to L8–L10 across most of mid-training (returning briefly to L12 at the step-16k dip), and converges to L10. The 1.4B starts at L17–L18, drifts to L10–L13 mid-training, and returns to L17 as the signal collapses.

Table 11: Checkpoint dynamics for two matched-width Pythia configurations ($d = 2048$). Both form healthy observability at the earliest measured checkpoint. The (16L/8H) configuration recovers after a mid-training dip; the (24L/16H) configuration does not, converging near the detection floor despite comparable perplexity.

Model	Config	Step	Tokens	Ppl	ρ_{partial}	$\pm$ std	r_{OC}	Peak	τ_{seed}
Pythia 1B	16L/8H	256	537M	1424.9	0.315	0.001	0.079	L12 (75%)	0.979
Pythia 1B	16L/8H	1k	2.1B	139.4	0.303	0.003	0.099	L12 (75%)	0.970
Pythia 1B	16L/8H	2k	4.2B	62.6	0.288	0.001	0.119	L10 (62%)	0.968
Pythia 1B	16L/8H	4k	8.4B	41.0	0.226	0.006	0.097	L8 (50%)	0.951
Pythia 1B	16L/8H	8k	16.8B	32.3	0.242	0.013	0.094	L10 (62%)	0.939
Pythia 1B	16L/8H	16k	33.6B	27.8	0.134	0.008	0.023	L12 (75%)	0.887
Pythia 1B	16L/8H	32k	67.1B	25.1	0.252	0.017	0.122	L10 (62%)	0.932
Pythia 1B	16L/8H	64k	134.2B	22.9	0.257	0.015	0.133	L10 (62%)	0.951
Pythia 1B	16L/8H	128k	268.4B	20.4	0.206	0.012	0.085	L10 (62%)	0.967
Pythia 1B	16L/8H	143k	299.9B	20.6	0.246	0.012	0.120	L10 (62%)	0.974
Pythia 1.4B	24L/16H	256	537M	1660.9	0.297	0.002	0.065	L17 (71%)	0.973
Pythia 1.4B	24L/16H	1k	2.1B	136.3	0.314	0.001	0.113	L18 (75%)	0.977
Pythia 1.4B	24L/16H	2k	4.2B	61.8	0.231	0.002	0.057	L13 (54%)	0.970
Pythia 1.4B	24L/16H	4k	8.4B	40.3	0.254	0.004	0.098	L10 (42%)	0.963
Pythia 1.4B	24L/16H	8k	16.8B	30.4	0.156	0.007	0.037	L11 (46%)	0.940
Pythia 1.4B	24L/16H	16k	33.6B	25.4	0.123	0.008	0.019	L13 (54%)	0.914
Pythia 1.4B	24L/16H	32k	67.1B	22.5	0.113	0.003	0.011	L17 (71%)	0.893
Pythia 1.4B	24L/16H	64k	134.2B	20.6	0.214	0.007	0.092	L13 (54%)	0.942
Pythia 1.4B	24L/16H	128k	268.4B	18.4	0.125	0.004	0.013	L17 (71%)	0.949
Pythia 1.4B	24L/16H	143k	299.9B	18.4	0.106	0.006	0.003	L17 (71%)	0.947

F TOKEN BUDGET SENSITIVITY

Cross-scale comparisons require matching token budgets to hidden dimension. Without this, larger models appear to have weaker signal because their higher-dimensional representations need proportionally more training examples.

At Qwen 2.5 0.5B ($d = 896$), a seven-point sweep from 150 to 1,000 examples per hidden dimension (ex/dim) reveals a detection threshold between 450 and 600 ex/dim (Figure 8). Below 450, ρ_{partial} is flat at 0.145 to 0.153 (7-seed means). At 150 ex/dim the seed std is 0.005, so the measurement is precise; the signal is weak because the probe is under-resourced, not because it is noisy. At 600, ρ_{partial} rises to 0.215, the value reported in Table 6, and continues to 0.232 at 800 and 0.248 at 1,000.

The threshold implies that small models ($d < 1,000$) require more data per dimension than larger models do to produce stable probes. All cross-scale comparisons in this paper use ex/dim ratios above the detection threshold for each model’s hidden dimension. An earlier measurement of Qwen 2.5 0.5B at 190 ex/dim (0.134) was below the detection threshold. The cross-family table reports the 600 ex/dim result (0.215), which clears the detection threshold; the endpoint remains mildly budget-sensitive between 600 and 1,000 ex/dim (0.215 to 0.248).

On Qwen 2.5 14B, increasing from 68 to 350 ex/dim on the same model raises ρ_{partial} from 0.194 to 0.214.

The ex/dim metric scales the training set with hidden dimension, so absolute token budgets grow with model size. At the reported ratios, the smallest cross-family model (GPT-2 124M, $d = 768$) uses $\sim 270,000$ training tokens and the largest hidden dimension ($d = 5120$, used by Qwen 2.5 14B and 32B) uses $\sim 1.8\text{M}$. Collapsed models are not bottlenecked by absolute budget: Llama 3B (0.091) and 8B (0.093) both run with absolute budgets above 10^6 tokens and still collapse. The within-family comparisons (Qwen 2.5 0.5B–32B, Llama 1B–8B, Pythia 70M–12B) hold ex/dim above each model’s detection threshold. The threshold’s dependence on absolute budget is out of scope.

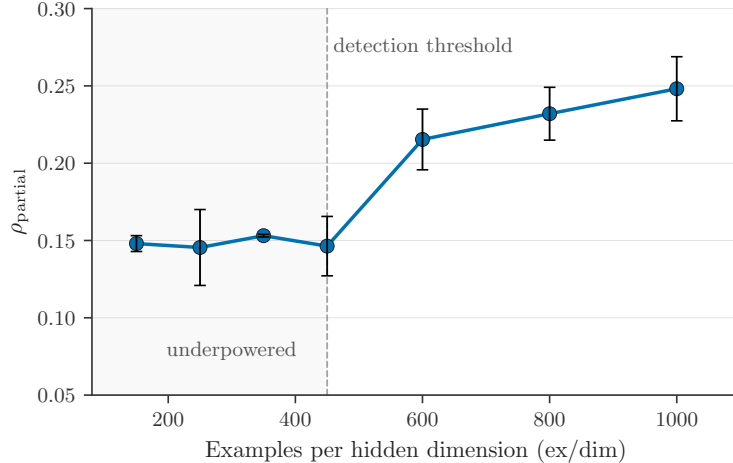


Figure 8: Ex/dim sensitivity at Qwen 2.5 0.5B. Seven token budgets showing ρ_{partial} (7-seed mean) at peak layer. The signal is below the detection threshold between 150 and 450 ex/dim and rises sharply between 450 and 600.

G CROSS-DOMAIN TRANSFER

G.1 THE PROBE IS PORTABLE BUT THE CALIBRATION DATA IS NOT

A WikiText-trained observer transfers to C4 web text across five families (Qwen, Llama, Gemma, Mistral, Phi), retaining 45–75% of the WikiText signal (Table 12). The probe reads the same kind of quality signal on text it was never trained on.

Table 12: Cross-domain transfer: WikiText-trained probe evaluated on C4 (transfer) vs. probe trained directly on C4 (within-domain). Percentages show C4 transfer as a fraction of WikiText signal.

Model	WikiText ρ_{partial}	C4 transfer	C4 within-domain
Qwen 2.5 3B	0.263	0.190 (72%)	−0.021
Qwen 2.5 7B	0.255	0.191 (75%)	−0.029
Llama 1B	0.286	0.159 (55%)	−0.046
Llama 3B	0.091	0.041 (45%)	−0.033
Mistral 7B	0.313	0.155 (50%)	−0.010
Phi-3 Mini	0.300	0.192 (64%)	0.223
Gemma 3 1B	0.216	0.151 (70%)	0.047

Percentages computed from unrounded values.

The asymmetry is consistent: WikiText→C4 transfer works, but C4→C4 within-domain training fails on most tested models, with Phi-3 Mini the exception (0.223). The probe is portable across distributions, but training requires calibrated data. C4 is noisier and more heterogeneous than WikiText, so the OLS residualization against confidence produces a noisier binary target. On most families the probe does not converge on a stable direction under these conditions, even with the same total token count. We do not have a predictor for which families succeed on within-domain C4 training.

G.2 CODE TRANSFER

On GPT-2 124M, a WikiText-trained probe transfers to CodeSearchNet (Husain et al., 2019) Python at $\rho_{\text{partial}} = 0.539$, exceeding the WikiText source signal (0.288). Transfer to OpenWebText (Gokaslan and Cohen, 2019) is weak (0.086), weaker than the C4 transfer pattern. The code result suggests the decision-quality signal encodes structural prediction difficulty (closing brackets, completing function signatures, matching indentation) that generalizes across natural language and code. Code has cleaner syntactic structure than web text, which may explain why the probe finds more signal there than on its training distribution.

G.3 LOW-OBSERVABILITY MODELS

Llama 3B shows weak signal on both WikiText (0.091) and C4 transfer (0.041). Llama 1B shows the same asymmetry as high-observability models: WikiText 0.286, C4 transfer 0.159, C4 within-domain -0.046 . Low observability at 3B is a property of the model’s representations, not the input domain.

H DOWNSTREAM EVALUATION

H.1 TASKS AND PROTOCOL

The same 20% ceiling recurs in seven of nine cells across three downstream tasks: SQuAD 2.0 (Rajpurkar et al., 2018), MedQA-USMLE (Jin et al., 2021), and TruthfulQA (Lin et al., 2022), measured on three production instruct models (Qwen 2.5 7B Instruct, Mistral 7B Instruct v0.3, Phi-3 Mini Instruct), each using its own WikiText-trained probe applied without task-specific training (Table 13). SQuAD 2.0 uses the first $n = 1,000$ answerable questions in dataset order from the validation split with provided context (no retrieval); MedQA-USMLE uses the first $n = 1,000$ items in dataset order from the four-option test split scored by exact option match; TruthfulQA uses the multiple-choice validation split (817 questions, MC1 single-correct format) scored by normalized substring match against the MC1 correct option. Each model generates an answer via greedy decoding with a minimal task prompt. The observer score and max-softmax confidence are averaged over generated tokens (excluding the prompt) to produce per-question scores. Questions are ranked by each score and flagged at the specified rate. A question is an exclusive observer catch if the observer flags it, confidence does not, and the answer is wrong.

Table 13: Exclusive catch rates across three production instruct models and three downstream tasks, using each model’s WikiText-trained probe applied without task-specific training. Columns report catch rate at 10% and 20% flag rate (percentage of errors caught exclusively by the observer). Seven of nine cells at 20% fall between 10.9% and 13.4%; the two exceptions (Phi-3 Mini Instruct on SQuAD at 6.6% and on MedQA at 4.1%) are task-model interactions discussed below. Confidence has higher single-signal precision at every flag rate; these are the observer’s non-overlapping catches.

Task	Qwen 2.5 7B-I		Mistral 7B-I		Phi-3 Mini-I	
	@10%	@20%	@10%	@20%	@10%	@20%
SQuAD 2.0	5.0%	10.9%	9.7%	11.0%	5.8%	6.6%
MedQA-USMLE	9.4%	13.4%	6.5%	11.2%	1.0%	4.1%
TruthfulQA	7.9%	12.7%	6.7%	10.9%	7.7%	13.1%

H.2 PER-CELL BREAKDOWN

Seven of the nine model-task cells at 20% flag rate fall between 10.9% and 13.4%, near the language-modeling ceiling band (12 – 15% at 20%, Table 7). Two cells sit below: Phi-3 Mini Instruct on SQuAD (6.6%) and MedQA (4.1%). Phi-3 Mini Instruct preserves observability on WikiText (0.300) and on TruthfulQA (13.1% at 20%), so the SQuAD and MedQA outcomes are task-model interactions rather than a floor on the method. The observer direction learned from Wikipedia next-token prediction reads the same signal across reading comprehension, medical, and factual QA (Figure 7). On MedQA with Qwen 2.5 7B Instruct, among the 40% of questions the model gets wrong, at 20% flag rate about one in seven is confidently wrong and caught only by the observer.

Random-baseline note. A task-specific random-ranker baseline analogous to the GPT-2 language-modeling baseline (Section 5) is not computed for the downstream cells in this paper. The language-modeling analog is 8.1% at 10% flag rate and 13.0% at 20%, derived from the disagreement of two rankers with fixed flag rates. The downstream catch rates above (10.9–13.4% at 20%) sit near this analog, so the operational claim is the non-overlapping catch of a frozen WikiText probe, not a margin over random. Computing a downstream-specific random-ranker baseline is left for future work.

H.3 TRUTHFULQA BOUNDARY

The catches above operate over all errors at a fixed flag rate. Among the narrower subset of confidently wrong answers, where the model asserts a fluent falsehood with high confidence, the observer does not discriminate (area under the ROC curve, AUC 0.499 on Qwen 2.5 7B-I, 0.568 on Mistral 7B-I,

Table 14: Base vs. instruct across Qwen 2.5 scales plus Mistral 7B. Instruction tuning preserves observability; changes are small relative to the cross-family spread.

Scale	Base ρ_{partial}	Instruct ρ_{partial}	Δ
Qwen 2.5 0.5B	0.215	0.312	0.097
Qwen 2.5 1.5B	0.275	0.266	-0.009
Qwen 2.5 3B	0.263	0.265	0.002
Qwen 2.5 7B	0.255	0.291	0.035
Qwen 2.5 14B	0.214	0.237	0.023
Mistral 7B	0.313	0.339	0.026

Deltas computed from unrounded values.

0.556 on Phi-3 Mini-I, all near chance). The observer catches token-level prediction failures at the aggregate rate; it does not catch fluent reproduction of learned falsehoods, and the three-model consistency defines that boundary as a property of the method, not of the model.

I INSTRUCTION-TUNING ROBUSTNESS

The cross-family pattern holds at every Qwen 2.5 scale tested (Table 14):

At 0.5B, instruction tuning increases the signal by $\Delta = 0.097$. At 1.5B through 14B, the delta is small (-0.009 to 0.035). The pattern extends beyond Qwen 2.5: Llama 3.2 1B Instruct produces 0.285 versus base 0.286 ($\Delta = -0.001$); Mistral 7B Instruct gives 0.339 against base 0.313 ($\Delta = 0.026$). Whatever produces observability runs deeper than what instruction tuning modifies.